COULSON AVIATION

LAT AIR ATTACK OPERATIONS GUIDE

COULSON AVIATION AUSTRALIA

Purpose

This guide has been developed to provide a clear framework for the conduct of Large Air Tanker Aerial Supervision operations within Coulson Aviation Australia. Its purpose is to establish consistent standards of practice that reflect both the complexity of the mission and the requirements of effective coordination with Airtanker crews.

This guide is based on multiple authoritative sources. These include the *NWCG Standards for Aerial Supervision,* the *CalFire Standards for Aerial Supervision*, the *NSW RFS Airtanker Operations Guideline,* and the *NSW and ACT Aviation Standard Operating Procedures*. Where applicable, NSW RFS policy has been incorporated to ensure alignment with local requirements.

This guide prioritises the NWCG and CalFire models, which represent robust and globally recognised standards for Aerial Supervision. These frameworks form the basis of Coulson's approach worldwide and provide a foundation for consistent, professional integration between pilots across diverse operational environments.

This guide reflects Coulson Aviation's commitment to those standards. It is intended to provide the clarity and structure necessary for safe, effective, and professional operations, ensuring Aerial Supervision is conducted in a manner consistent with best practice and global expectations. As operational experience is gained and standards continue to evolve, this guide will be reviewed and updated to remain current, relevant, and aligned with both organisational and industry development.

Record of Revisions

Version	Date	Details
V1.0	November 2025	Initial Release

Note: Changes made in the current version are annotated with change bars.

Table of Contents

1 ACRONYMS AND DEFINITIONS 1

1.1 List of Acronyms 1
1.2 Key Operational Terms 2

2 AERIAL SUPERVISION: ROLES, MISSIONS, AND RESPONSIBILITIES 7

2.1 The Air Attack Supervisor (AAS) - Foundational Role 7
2.2 Fixed-Wing Missions 7
2.3 Helicopter Missions 8
2.4 Pilot Roles and Qualifications 9
2.5 Aerial Supervision Organisation 9

3 SAFETY AND RISK MANAGEMENT 10

3.1 Core Safety Principles 10
3.2 Flight Risk Assessment Tool (FRAT) 10
3.3 Dynamic Risk Assessment 10
3.4 Refusal of Unsafe Assignments – Task Rejection 11
3.5 Incident Reporting 11

4 INCIDENT AIRCRAFT 13

4.1 Aircraft Typing and Classification 13
4.2 Airtanker Typing 13
4.3 Airtanker Capabilities 13
4.4 Rotary-Wing Typing 14
4.5 NAFC National Fleet 14
4.6 NSW RFS Fleet 14

5 POLICIES, REGULATIONS, AND GUIDELINES 16

5.1 Aerial Supervision Requirements 16
5.2 Personal Protective Equipment (PPE) 17
5.3 Duty and Flight Time Limitations 17
5.4 Incident Airspace; the Fire Traffic Area 19

6 MISSION PROCEDURES 21

6.1 Aircraft Preparation 21
6.2 Flight Risk Assessment Tool (FRAT) 21
6.3 Dispatch Procedures 21
6.4 En Route Procedures 22
6.5 Fire Traffic Area Entry Procedures 23
6.6 Entering Incident Airspace 24
6.7 First Aerial Supervision on Scene – Immediate Responsibilities 24
6.8 Fixed-Wing Scripts 24
6.9 Standard Clearances and Commands 24
6.10 Air Attack Flight Profiles 25
6.11 Maintaining Continuity of Supervision 27
6.12 Transition Briefings 27
6.13 Non-Normal Situations 28

7 FIRE TRAFFIC AREA MANAGEMENT 30

7.1 Common Principles of Aircraft Separation 30
7.2 Vertical Separation 30
7.3 Horizontal Separation 31
7.4 Working Area/Area of Operation (WA/AO) 32
7.5 Active Sequencing 33
7.6 Firebombing and CTAF Frequency Management 35

7.7 Coordination Between Aerial Supervisors 35

8 FIRE BEHAVIOUR AND FIRE WEATHER 36

8.1 Fundamentals of Fire Behaviour 36
8.2 The Influence of Fire Weather 37
8.3 Types of Fire Activity 38

9 AERIAL FIRE SUPPRESSION STRATEGIES 39

9.1 Aerial Fire Suppression Strategies 39
9.2 Aerial Fire Suppression Tactics 39
9.3 IA and Multiple Fire Operations 40
9.4 Delayed Attack Fires 41
9.5 Wildland Urban Interface 41
9.6 Fire Chemicals 42
9.7 Coverage Level Section 43
9.8 Ten Principles of Retardant Application 44

10 SITUATIONAL AWARENESS AND DECISION-MAKING 46

10.1 Maintaining Situational Awareness 46
10.2 Watch Out Situations 47
10.3 Judgment in High-Risk Environments 47
10.4 Dynamic Risk Assessment During Operations 48
10.5 The Eighteen Watch Out Situations 49

11 POST-MISSION RESPONSIBILITIES 50

11.1 After-Action Review (AAR) 50
11.2 Administrative Duties 50
11.3 Safety Reporting 50

APPENDIX A – FIXED-WING SCRIPTS I

1 Acronyms and Definitions

1.1 List of Acronyms

Acronym	Definition
AAR	After-Action Review
AAS	Air Attack Supervisor
ABM	Air Base Manager
ACT	Australian Capital Territory
AGL	Above Ground Level
ALARP	As Low as Reasonably Practicable
AMSL	Above Mean Sea Level
AO	Area of Operation
ARFF	Aircraft Rescue and Fire Fighting
ATSB	Australian Transport Safety Bureau
BDOG	Birddog
BMBR	Bomber
CASR	Civil Aviation Safety Regulations
CRM	Crew Resource Management
CTAF	Common Traffic Advisory Frequency
DFO	Director of Flight Operations
FBIR	Firebird
FCTAF	Fire Common Traffic Area Frequency
FRAT	Flight Risk Assessment Tool
FTA	Fire Traffic Area
FSCN	Firescan
IA	Initial Attack
HLTK	Helitak
HOFO	Head of Flying Operations
IAAS	Incident Air Attack Supervisor
IAP	Incident Action Plan
IC	Incident Controller / Incident Commander
ICS	Incident Command System
KIAS	Knots Indicated Airspeed
LAT	Large Airtanker

LAT AAS	Large Airtanker Air Attack Supervisor
LAT Tactical Frequency	The VHF-AM frequency used by Large Airtankers and aerial supervision to conduct tactical operations.
MSL	Mean Sea Level
NM	Nautical Mile
NSW	New South Wales
NWCG	National Wildfire Coordinating Group
PPE	Personal Protective Equipment
RFS	Rural Fire Service
SAD	State Air Desk
SAR	Search and Rescue
SEAT	Single Engine Air Tanker
SMS	Safety Management System
SOP	Standard Operating Procedure
SPTR	Firespotter
V_{MCA}	Minimum Controllable Airspeed (with critical engine inoperative)
WA	Working Area
WUI	Wildland Urban Interface

1.2 Key Operational Terms

Term	Definition
Active Sequencing	The process of coordinating and directing multiple aircraft through the Fire Traffic Area in a safe and efficient sequence.
Aerial Supervision	The generic term for the airborne command, control, and coordination of aerial resources over an incident. This function is performed during either an Air Attack Mission or a Leadplane Mission.
Air Attack Supervisor	An aerial supervisor qualified to coordinate all aerial resources at an incident, typically operating from a fixed-wing aircraft or helicopter.
Air Attack	The colloquial term for an Air Attack Supervisor (AAS).
Air Attack Pilot	A pilot qualified to fly fixed-wing supervision missions.
Airtanker	A fixed-wing aircraft equipped to deliver fire retardant, gel, or water to suppress wildfires.
Airtanker Base	A location where airtankers are loaded with retardant, gel, or water and serviced for operations.

Airtanker Manoeuvring Altitude	The designated altitude at which airtankers operate while positioning for drops or exiting from the Fire Traffic Area.
Anchor Point	An advantageous location, usually a barrier to fire spread, from which to start building a fireline.
Avoidance Area	Designated airspace around critical infrastructure, populated areas, or hazards where aircraft should not operate.
Base	The flight leg at a right angle to the final approach, connecting the downwind leg to the final approach.
Chase Profile	A tactical flight profile where the Leadplane follows behind the airtanker to observe and assess the drop.
Check-In	The process by which aircraft report their arrival and readiness to the aerial supervisor.
Checkpoint	A defined geographic or visual reference used to sequence, hold, or report helicopter positions for coordinated traffic flow
Coverage Level	The concentration of retardant applied to the ground, measured in gallons per 100 square feet.
Direct Attack	A fire suppression method where control lines are constructed directly on the fire edge.
Dispatch	The process of ordering and releasing aircraft to/from an incident.
Downwind	A flight leg parallel to the drop line, flown in the opposite direction to the intended drop run.
Drop Altitude	The altitude at which retardant or water is released from an airtanker, measured in MSL
Drop Height	The vertical distance between the aircraft and the ground at the point of retardant or water release, measured in AGL.
Drop Pattern	The shape and distribution of retardant or water on the ground after release from an aircraft.
Escape Route	A preplanned route for personnel to move away from a fire to a safety zone.
Exit	A predetermined flight path route for aircraft to climb away from the target drop area,
Final Approach	The last leg of the flight path before the drop, aligned with the intended drop line.
Fire Behaviour	The manner in which a fire reacts to fuel, weather, and topography.
Fire Traffic Area (FTA)	The designated airspace around a wildfire incident where aerial firefighting operations are conducted.
Go-Around	An aborted drop approach where the aircraft climbs away without releasing retardant or water.
High-Level Assessment	An aerial evaluation conducted from altitude to assess the overall fire environment, terrain, wind, smoke, and hazards, and to identify

	suitable approach, exit, and holding areas prior to low-level operations.
Holding Pattern	A predetermined flight path used by Airtankers while waiting for clearance to enter the Fire Traffic Area or by helicopters while waiting for clearance to proceed beyond a checkpoint or virtual fence.
Incident Action Plan (IAP)	A comprehensive plan that outlines strategies, tactics, and resource assignments for managing an incident.
Incident Air Attack	An AAS who provides low-level supervision from a helicopter. See Section 2.1.
Indirect Attack	A fire suppression method where control lines are constructed away from the active fire edge.
Initial Attack (IA)	The first suppression action taken on a wildfire, aimed at controlling it while it is still small.
Join-Up	The procedure by which an airtanker establishes visual contact and positions itself relative to the Leadplane for coordinated operations.
LAT AAS	An AAS with a specialist qualification for directing Large Air Tankers. See Section 2.1.
LAT Working Area	A defined sector of airspace within the Fire Traffic Area dedicated to Large Airtanker operations.
Lead Profile	A tactical flight profile where the Leadplane flies ahead of the airtanker to mark the drop location and demonstrate the flight path.
Leadplane	An aircraft used to guide airtankers to the target, mark drop locations and provide tactical coordination.
Leadplane Pilot	An Air Attack Pilot who also holds an advanced qualification for low-level flight. See Section 2.1.
Line	A fireline or control line constructed to contain or control a fire.
Load and Return	The process of an airtanker returning to base, reloading, and returning to the incident.
Long-Term Retardant	Fire retardant that remains effective for extended periods, typically containing ammonium phosphate or sulphate.
Low-Level Operations	Flight operations conducted at altitudes below 500 feet AGL.
Low-Level Recon	A reconnaissance flight conducted at operational altitude to confirm approach and exit routes, drop parameters, target features, escape routes, visibility, and to identify hazards not visible from higher altitude.
Airtanker Orbit Altitude	The altitude, typically 1,500 feet AGL, that provides safe vertical separation from other aircraft while the airtanker holds awaiting clearance to the target.
Parallel Attack	A fire suppression method where control lines are constructed roughly parallel to but some distance from the fire edge.

Positive Control	A method of airspace management where the aerial supervisor maintains direct communication and authorisation for all aircraft movements.
Reconnaissance	Aerial observation to gather information about fire behaviour, location, and suppression needs.
Release	The authorisation for an aircraft to depart the incident when it is no longer required by the fire operation
Safety Zone	An area cleared of flammable materials where firefighters can take refuge from a fire.
Safety Manager	Responsible for the implementation and continuous operation of the organisation's Safety Management System (SMS).
Safety Management System (SMS)	A structured approach for identifying hazards, assessing the risk to safety, and implementing measures to maintain safe operations.
Show-Me	A tactical flight profile where the Leadplane demonstrates the intended drop run without the airtanker following immediately behind.
Single Engine Air Tanker (SEAT)	A small, single-engine aircraft equipped to deliver fire retardant, gel, or water.
Span of Control	The number of aircraft or resources that can be effectively supervised by a single individual without compromising safety or operational effectiveness.
Spot Fire	A fire ignited outside the main fire perimeter by flying embers or sparks.
Staging Area	A location where aircraft wait for assignment or clearance to enter the Fire Traffic Area.
Task Saturation	A condition in which the number or complexity of tasks exceeds a pilot's capacity to process information or respond effectively, often leading to reduced situational awareness and impaired decision-making.
Tactical Pause	A temporary suspension of operations to reassess conditions, address safety concerns, or adjust tactics.
Target	The specific location where retardant or water is intended to be dropped.
Torching	The ignition and rapid combustion of aerial fuels, usually trees or shrubs.
Transition Briefing	Information exchange between incoming and outgoing aerial supervisors to ensure continuity of operations.
Turn Time	The elapsed time from when an aircraft departs the incident to when it returns ready for another drop.
Virtual Fence	A defined boundary used to delineate separate working areas and maintain safe separation between aircraft, including between helicopters or between helicopters and fixed-wing operations.

Visual Flight Rules (VFR)	Regulations that allow pilots to operate aircraft in weather conditions clear enough to see where they are going.
Waterway	A body of water (lake, river, reservoir) used by water scoopers for loading operations.
Working Area (WA)	A designated area of airspace assigned to specific aircraft or operations to maintain separation and coordination.

2 AERIAL SUPERVISION: ROLES, MISSIONS, AND RESPONSIBILITIES

2.1 The Air Attack Supervisor (AAS) - Foundational Role

The Air Attack Supervisor (AAS) is the foundational qualification for all airborne coordinators in the aerial firefighting environment. The AAS is the designated single point of contact for all aircraft operating at an incident and is the crucial link between the air resources and the Incident Controller on the ground. The responsibilities listed below are the core duties of any AAS, regardless of the mission or aircraft platform. Specific duties for different missions are detailed in the following sections.

Core Responsibilities:

- Supervise all aerial resources assigned to the incident.
- Ensure alignment between aerial operations and ground objectives.
- Monitor and mitigate risks associated with aerial operations.
- Provide coordination and oversight of all incident aircraft.

2.2 Fixed-Wing Missions

Fixed-wing aircraft serve as the primary platform for all operations involving Large Air Tankers (LATs). The specific mission flown is determined by the qualifications of the pilot crewing the aircraft with the LAT AAS.

2.2.1 LAT Air Attack Mission

Crew: LAT AAS + Air Attack Pilot

Profile: A mission conducted at an altitude appropriate for effective observation, coordination, and deconfliction of Large Airtanker (LAT) operations. The team orbits above the incident to provide tactical coordination and airspace management for all aerial assets, including LATs.

In this configuration, the LAT AAS maintains full control of LAT and helicopter operations within the LAT working area and conducts all communications on agency frequencies, including dispatch, FCTAF, and LAT tactical. The Air Attack Pilot provides flight management and aircraft handling support to facilitate effective supervision and communication.

Mission Core Functions:

- Provide strategic oversight and deconfliction for all aircraft operating in the LAT working area.
- Coordinate tasking and sequencing of LATs and supporting rotary and fixed-wing assets.
- Maintain overall airspace situational awareness.
- Advise incident command on aerial resourcing and safety.

LAT AAS Responsibilities:

- Coordinate and direct LAT operations and sequencing.
- Provide strategic guidance to the IAAS, Incident Controller and other relevant personnel regarding LAT deployment.

- Manage all aircraft operating within the LAT working area and ensure orderly integration of resources.
- Ensure compliance with safety standards and adapt operations as conditions evolve.

Air Attack Pilot Responsibilities:

- Operate the Air Attack aircraft safely and in accordance with agency and operator procedures.
- Maintain separation from terrain and other aircraft, complying with altitude and spacing requirements.
- Provide a stable, appropriately positioned observation platform to support the AAS's tactical oversight.
- Manage aircraft systems, navigation and communications required to support the mission.
- Maintain situational awareness of the FTA and report identified hazards or conflicts to the AAS.

2.2.2 Leadplane Mission

Crew: LAT AAS + Leadplane Pilot

Profile: A specialized, low-level mission where the team provides close-in, visual guidance for airtankers. The aircraft flies the intended drop path at low altitude ahead of the airtanker to demonstrate the route and mark the target.

In this configuration, the LAT AAS retains overall control of LAT tasking and communicates with dispatch and FCTAF, while the Leadplane Pilot conducts direct tactical communication with the airtankers on the LAT tactical frequency and provides low-level pilot guidance.

Mission Core Functions:

- Provide tactical guidance and supervision for LATs and helicopters within the LAT working area.
- Enhance operational safety through proactive risk management.
- Adapt dynamically to changes in fire behaviour and mission priorities.

LAT AAS Responsibilities:

- Coordinate and direct LAT operations.
- Provide strategic guidance to the IAAS, IC, and other personnel on LAT deployment.
- Manage all aircraft operating within the LAT working area.
- Ensure compliance with safety standards and adapt operations as conditions evolve.

Leadplane Pilot Responsibilities:

- Operate the Leadplane safely and mitigate risks associated with low-level flight.
- Execute Leadplane profiles and manoeuvres to guide LATs.
- Assist with helicopter sequencing within the LAT working area as directed by the LAT AAS.
- Maintain situational awareness and adapt tactics to changing fire conditions.

2.3 Helicopter Missions

Helicopters provide a versatile platform for low-level aerial supervision, particularly for fires where LATs are not engaged.

2.3.1 Incident AAS

Crew: Incident AAS + Helicopter Pilot

Profile: The helicopter typically operates at low altitude, providing the AAS with a close-in view of the fireground and direct oversight of helicopter-based fire suppression and support operations.

Incident AAS Responsibilities:

- Supervise all aerial resources assigned to the incident.
- Ensure alignment between aerial operations and ground objectives.
- Monitor and mitigate risks associated with aerial operations.
- Provide coordination and oversight of all incident aircraft.

2.4 Pilot Roles and Qualifications

The qualification of the pilot determines the mission capability of the fixed-wing supervision team.

2.4.1 Air Attack Pilot

The Air Attack pilot is qualified to command the fixed-wing supervision aircraft. This qualification allows them to conduct LAT Air Attack Missions in conjunction with a LAT AAS.

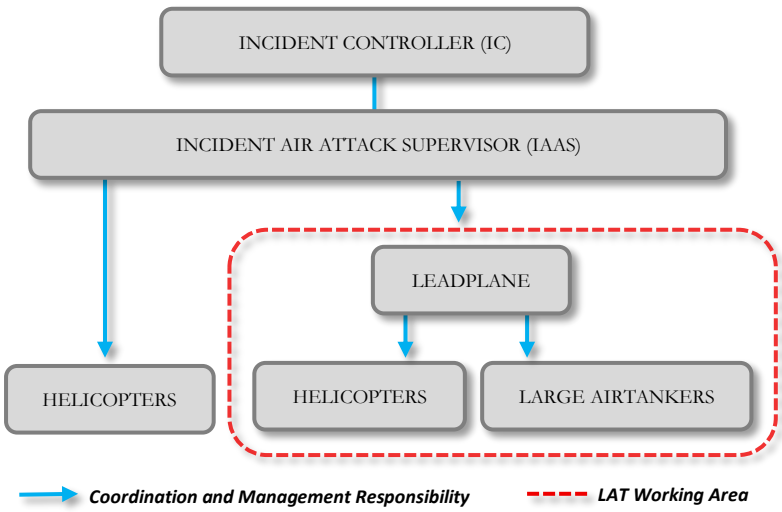
2.4.2 Leadplane Pilot

The Leadplane pilot has completed an advanced training syllabus to gain the specialized skills required to provide low-level tactical guidance to Airtankers in the fire environment. This additional qualification is required to conduct Leadplane Missions.

2.5 Aerial Supervision Organisation

When an IAAS is present, the IAAS oversees all incident aircraft and ensure alignment with incident objectives. When a Leadplane is activated, the LAT AAS assumes responsibility for managing aircraft within the LAT working area, a defined sector of airspace dedicated to LAT operations.[1]

Figure 1. Aerial Supervision Organisation – Initial Attack



[1] In the absence of an Incident AAS, the LAT AAS may also assume the IAAS function.

3 Safety and Risk Management

3.1 Core Safety Principles

Safety is the cornerstone of all operations, ensuring the protection of personnel, equipment, and the environment. Coulson Aviation prioritises a proactive and collaborative approach to safety to support effective decision-making and operational success.

Core safety principles include

- *Proactive Risk Mitigation*—Identify and address potential hazards before they escalate into operational risks.
- *Adaptability*—Adjust strategies and actions dynamically to respond to changing conditions in real time.
- *Crew Resource Management (CRM)*—Leverage the collective knowledge, expertise, and perspectives of all team members to enhance decision-making and operational safety.
- *Standardisation*—Adhere to established protocols and procedures to maintain consistency and minimise variability in safety practices.

3.2 Flight Risk Assessment Tool (FRAT)

The FRAT is a structured evaluation designed to help identify, assess, and mitigate risks to a level that is As Low as Reasonably Practicable (ALARP), meaning risks are reduced to the lowest level achievable considering cost and feasibility. It provides a standardised method for assessing factors such as crew readiness, mission complexity, and environmental conditions, and is intended as a decision-support tool rather than a compliance exercise.

Pilots are expected to complete a FRAT before the Daily Strategic Briefing and save it in a pending status. This allows for adjustments based on updated mission details or briefing outcomes. Once dispatched, the FRAT must be reviewed and finalised. If conditions change during the day, the FRAT should be updated between missions to ensure the risk profile remains accurate.

The FRAT must also be completed and saved at the end of the day, even if no flights occurred, as it provides an essential record for monitoring the organisation's overall risk profile.

3.2.1 Interpreting FRAT scores

- *Green*—Acceptable Risk—proceed with mission.
- *Yellow*— Elevated Risk—approval from DFO, HOFO, or Safety Manager is required. Non-essential crewmembers or passengers (e.g., engineers) are not permitted on the aircraft.
- *Red*—Unacceptable Risk—mission not authorised unless risks can be mitigated to an acceptable level. If not, the mission must be cancelled.

3.3 Dynamic Risk Assessment

Dynamic risk assessment involves the continuous evaluation of hazards and operational conditions throughout the mission. This process is critical in responding to the ever-changing nature of aerial firefighting environments and requires a high degree of situational awareness.

Key elements of dynamic risk assessment include

- *Situational Awareness*—Pilots and supervisors must remain vigilant, identifying and responding to changes in fire behaviour, weather, and airspace congestion in real time.
- *Communication and Collaboration*—CRM principles are vital to dynamic risk assessment, enabling the team to draw upon the collective knowledge and expertise of all personnel involved. Open communication ensures that potential hazards are identified early and mitigated effectively.
- *Adaptability*—Operations must remain flexible to accommodate rapid changes, such as unexpected shifts in fire behaviour, evolving weather conditions, or the presence of additional aircraft.

3.4 Refusal of Unsafe Assignments – Task Rejection

The refusal of unsafe assignments is a critical safety mechanism that empowers personnel to prioritise safety over operational objectives when risks are deemed unacceptable. Coulson Aviation fosters a safety culture that encourages informed and confident decision-making in these scenarios.

Flight crews are required to follow the New South Wales Rural Fire Service (NSW RFS) task rejection protocol when declining assignments. This ensures refusals are communicated clearly, documented in accordance with agency requirements, and accompanied by a *Prism Task Rejection Safety Report* to formally record the circumstances and rationale for the decision.

3.4.1 Guidelines for task rejection

- *Clear Communication*—The individual refusing the assignment must articulate their concerns and the reasons for refusal to the AAS, IAAS, or ABM as appropriate.
- *Propose Alternatives*—Whenever possible, safer alternative approaches should be offered to achieve the mission objective.
- *Documentation*— All task rejections must follow the NSW RFS task rejection protocol and include submission of a *Prism Task Rejection Safety Report* to support transparent communication and continuous improvement within the safety management system.

3.5 Incident Reporting

Incident reporting is a vital component of Coulson Aviation's safety program, enabling the identification of hazards, sharing of lessons learned, and continual improvement of safety practices.

Coulson Aviation flight crews are required to submit safety reports through the Prism SMS safety reporting system. This system ensures that safety events, including incidents and near-misses, are documented and addressed promptly.

What to Report

Any event that deviates from standard operating procedures, causes or has the potential to cause harm, or highlights a gap in safety measures.

How to Report

Use the Prism SMS system to submit detailed accounts of the event, including contributing factors, corrective actions, and suggested mitigations.

Post-Reporting Actions

All reports are reviewed, and findings are integrated into training programs and operational updates to enhance safety practices and reduce recurrence.

4 Incident Aircraft

Aerial Supervisors shall maintain an understanding of the aircraft types under their supervision, including communication requirements, performance characteristics, and logistical support considerations.

In Australia, tactical and logistical aircraft may be sourced through the National Aerial Firefighting Centre (NAFC), the New South Wales Rural Fire Service (NSW RFS), other state and territory emergency service agencies, or contracted commercial operators. Contractual and operational standards may vary between jurisdictions based on procurement arrangements.

4.1 Aircraft Typing and Classification

Aerial firefighting aircraft are categorised according to their role and capability as defined by the National Aerial Firefighting Centre (NAFC). These classifications describe each aircraft's primary function—such as Large Airtanker, Single Engine Airtanker, Rotary-Wing Suppression, or Reconnaissance—and outline key performance parameters including payload capacity, endurance, and equipment fit-out.

Aircraft capability categories support consistent tasking, coordination, and resourcing across jurisdictions. Current NAFC standards and capability definitions are detailed in *NAFC Standard OPS-006: Aircraft Capability and Category Definitions*[2].

4.2 Airtanker Typing

Fixed-Wing aircraft used for firebombing are assigned a type based on water carrying capacity:

Type	Engines	Descriptor	Capacity
FW 1	Multi Engine	Large Airtanker	Greater Than 11,356 Litres
FW 2	Multi Engine	Medium Multi Engine Air Tanker	Between 6,813 and 11,356 Litres
FW 3	Multi Engine	Small Multi Engine Air Tanker	Less Than 6,813 Litres
FW 4	Single Engine	Medium Single Engine Air Tanker	Greater Than 2,270 Litres
FW 5	Single Engine	Small Single Engine Air Tanker	Less Than or Equal to 2,270 Litres

4.3 Airtanker Capabilities

Type	Aircraft Make and Model	Maximum Litres	Cruise Speed (Knots)	Tank System
FW 1	B-737	15,140	450	Constant Flow
FW 1	C-130H/Q	15,140	300	Constant Flow
FW 1	RJ-85	11,355	340	Constant Flow
FW 2	Q-400	9,841	320	Constant Flow
FW 4	Air Tractor AT-802 F	3,028	170	Constant Flow

[2] NAFC Standard OPS-006: Aircraft Capability and Category Definitions.

Constant Flow System – A single compartment with two doors controlled by a computer. The system is capable of single or multiple even flow drops at designated coverage levels from.5 gallons per 100 square feet (GPC) to +8 GPC.

4.4 Rotary-Wing Typing

Rotary-Wing aircraft capable of firebombing have a type based on internal payload and water carrying capacity. Different types may be specified for different configurations (e.g. an aircraft may be RW Type 1 when operating with a tank and RW Type 2 with a bucket).

Type	Internal Payload	Capacity
RW 1	2,268 kg or Greater	2,650 Litres or Greater
RW 2	Between 1,1235 kg and 2,267 kg	Between 1,135 and 2,649 Litres
RW 3	Between 544 kg and 1,133 kg	Between 380 and 1.134 Litres
RW 4	Less Than 544 kg	Less Than 380 Litres

4.5 NAFC National Fleet

NAFC contracts approximately 170 aircraft on behalf of state and territory governments. This fleet is supplemented by NAFC aircraft contracted as Call When Needed, and additional state owned and state contracted aircraft hired to meet peak demand across Australia. In total more than 500 aircraft, provided by over 150 operators, are available for firefighting across Australia.

For an overview of aircraft currently contracted for aerial firefighting in Australia, refer to the NAFC Fleet Information[3] page, which provides general details on aircraft types, capabilities, and seasonal deployment.

4.6 NSW RFS Fleet

In addition to the contract fleet, NSW Rural Fire Service owns aircraft to support firefighting and other emergency operations. These aircraft and their associated roles are summarised below:

Aircraft	Roles
Boeing 737	Firebombing and Transport (pending regulatory approval).
CH47 Chinook	Tanked firebombing operations (including night operations) and emergency equipment transport (pending regulatory approval).
Bell 412	Aerial intelligence, Remote Area Firefighting Teams (RAFT), Rapid Aerial Response Teams (RART), Aviation Rescue Operations (ARC), Search and Rescue (SAR), Flood Rescue Winching; Road Crash Rescue (RCR), Aerial Incendiary (AI), Firebombing and Transport.
Cessna Citation	Aerial intelligence, Leadplane/Air Attack Supervision (AAS) and Transport.
King Air 350	Aerial intelligence, Leadplane/Air Attack Supervision (AAS) and Transport.

[3] NAFC Fleet Information

4.6.1 Callsigns

Aircraft involved in firefighting and emergency operations are assigned Special Task Operations (STO) callsigns for use in radio communications, documentation, and operational systems.

STO callsigns serve to:

- Clearly identify firefighting aircraft within air traffic and agency systems;
- Indicate the nature of operations to Air Traffic Services;
- Support situational awareness and coordination across agencies; and
- Provide consistency with agency radio networks and procedures.

STO callsigns used for fire and emergency operations are defined in the Aeronautical Information Publication (AIP), published by Airservices Australia:

Flight ID	Callsign	Aircraft Type	Operation Type
BDOG	BIRDDOG	Fixed-Wing	Coordination of Firebombing Aircraft
BMBR	BOMBER	Fixed-Wing	Firebombing
FBIR	FIREBIRD	Rotary-Wing	General Fire Support Tasks (Light Rotary)
FSCN	FIRESCAN	Fixed-Wing	Remote Sensing Fire Operations
HLTK	HELITAK	Rotary-Wing	General Fire Support Tasks (Medium Rotary)
SPTR	FIRESPOTTER	Fixed-Wing	Fire intelligence Gathering

4.6.2 Callsign Awareness and Deconfliction

While callsign conventions are nationally recognised, variations exist between agencies. In New South Wales, the NSW RFS uses the callsign "*Firescan*" for its agency owned multi-role fixed-wing aircraft, including the Citation and King Air fleet. These aircraft may perform the Leadplane or Aerial Supervision role despite the callsign's traditional association with reconnaissance and high-altitude intelligence operations.

This may lead to ambiguity, as participating aircraft could reasonably assume that a Firescan aircraft is operating at altitude rather than in a low-level supervisory role. Such assumptions can affect situational awareness and separation planning within the Fire Traffic Area. To maintain clarity:

- NSW RFS aircraft assigned the "Firescan" callsign must use that callsign but should announce their operational role at initial check-in (e.g. "Firescan Two-Zero-One, a Citation operating as a low-level Leadplane").
- All crews should confirm the role of Firescan aircraft during initial coordination and remain alert for any callsign-related ambiguity.
- Coulson Aviation Citations operating under NAFC contracts are assigned the "Birddog" callsign and shall use it when performing the Leadplane role.

This guidance promotes clear communication, shared situational awareness, and safe integration of aircraft within the FTA.

5 Policies, Regulations, and Guidelines

5.1 Aerial Supervision Requirements

When aerial supervisors are co-located with retardant aircraft, they will be launched together on the initial order to maximise safety, effectiveness, and efficiency of incident operations. NSW RFS policy dictates additional requirements as listed below.[4]

Table 1. Incident Aerial Supervision Requirements

SITUATION	LEADPLANE	IAAS
Three or more aircraft over an incident or when mixed type and kind aircraft are over the incident working at the same time.	ORDERED IF NO IAAS AND ONLY FIXED-WING	ORDERED
Fixed-Wing Low-Level Operations in Low Light conditions.	REQUIRED IF NO IAAS	REQUIRED IF NO LEADPLANE
Airtanker not IA Rated	REQUIRED	N/A
Periods of marginal weather, poor visibility, or turbulence.	REQUIRED	N/A
When requested by Airtankers, Helicopters, AAS, or Leadplane.	REQUIRED	REQUIRED

Required

Aerial supervisory resource(s) shall be over the incident when specified air tactical operations are being conducted. Required aerial supervision resources shall be on scene before dispatching tactical resources (helicopters or Airtankers) during periods of marginal weather, poor visibility, or turbulence.

Ordered

Aerial supervisors shall be ordered by the unit maintaining operational control (operations may be continued while the aerial supervisor is enroute to the incident. Operations can be continued if the resource is not available and assigned resources are notified).

N/A

Not authorised or applicable to the level of supervision required for the mission/resource.

5.1.1 Low Light Conditions (Sunrise/Sunset)

Low-light conditions are defined as the period beginning one (1) hour prior to evening civil twilight (ECT). During this period, natural illumination begins to decrease, reducing visibility and terrain contrast.

[4] Deviations from this table may be authorised by the agencies through local mitigations.

Low-level fixed-wing operations must cease no later than thirty (30) minutes prior to evening civil twilight and may not commence earlier than thirty (30) minutes after morning civil twilight.

Aerial supervision is required for all aerial firefighting operations conducted during low-light conditions.

5.1.2 Visibility

Visibility must meet the CASA Visual Meteorological Conditions (VMC) minima for the class of airspace in which operations are conducted. When visibility is insufficient to ensure safe operations, flights shall be suspended. All incident aircraft are strongly encouraged to operate with external lights on, appropriate to the aircraft and prevailing conditions. Regular position reporting is essential in marginal visibility to maintain situational awareness and flight safety.

5.1.3 Hazardous Conditions

Moderate to high winds and turbulent conditions affect flight safety and water/retardant drop effectiveness. Several factors including terrain, fuel type, target location, resources at risk, crosswinds, etc., must be considered. Aerial operations should cease when safety-of-flight is or may be compromised, water/retardant drops become ineffective, or at the pilot's recommendation.

Hazardous conditions may include, but are not limited to:

- Surface visibility less than 3 NM
- Cloud base lower than 2,000 ft AGL over the incident
- Sustained surface winds greater than 25 kt
- Forecast or observed moderate or greater turbulence below 3,000 ft AGL
- Thunderstorm activity within ~15 NM

Evaluate thunderstorms and other hazardous weather activities for flight safety. Erratic winds, lightning, hail, and diminished visibility adversely affect aviation operations. Consider delaying operations or reassigning resources to safe operation areas. Suspend flight operations when lightning or other adverse weather conditions are present.[5]

5.2 Personal Protective Equipment (PPE)

The following PPE is required during tactical operations (AAS and Pilot):

- *Flight Suit*—One-piece fire-resistant polyamide or aramid material or equal. The use of Nomex® shirts and trousers (two-piece) is authorised.
- *Protective Footgear*— Boots shall provide above-ankle support and be constructed of durable, fire-resistant or heat-tolerant materials.

5.3 Duty and Flight Time Limitations

Flight and duty limitations shall comply with:

- Civil Aviation Safety Regulations (CASA) Part 48.1;
- NSW RFS *Airtanker Operations Guideline*; and
- Coulson Aviation's Fatigue Management System.

[5] Any aerial supervisor, pilot, or ground resource can halt operations to mitigate risk or hazardous situations.

5.3.1 Day/Night Flight Policy

Multi-engine fixed-wing aircraft may transit to or from an incident, or operate over an incident, before sunrise or after sunset provided the aircraft and pilot are equipped and authorised for Instrument Flight Rules (IFR) operations, and all requirements for low-level flight in reduced-light conditions outlined in this guide are met. Refer to agency policy for any additional restrictions or clarifications.

5.3.2 Flight Following

A frequency is assigned by the dispatch centre for check-ins and incident related information. Dispatch centres may require a 30-minute check-in or a confirmation that an aircraft is showing "positive" on the automated flight following (SkyTrac) system.[6]

To reduce the amount of radio traffic and workload, the LAT AAS should carry SAR for all Airtankers under their control.

5.3.3 Air-to-Ground Communications

A dedicated air-to-ground frequency must be established and continuously monitored by aerial supervision resources.

- *Initial Attack Incidents*—Many agencies have pre-assigned FM air-to-ground frequencies assigned to geographic areas. Other agencies use standard work channel frequencies.
- *Extended Attack Incidents* — Specific frequencies should be requested through the dispatch system to avoid interference with other incidents. Some incidents may require two separate air-to-ground frequencies to distinguish between command and tactical communications. Once assigned, all incident frequencies and their designated purposes will be documented in the Air Operations Plan and the Incident Communications Plan, consistent with AIIMS and NSW RFS procedures.

5.3.4 Air-to-Air Communications

Communication between all airborne incident aircraft is critical to safety and effectiveness. Air-to-air communications are usually accomplished using a VHF-AM frequency. For tactical use, a Fire CTAF (FCTAF) and a LAT tactical frequency will be assigned to each incident.[7]

- FCTAF frequencies are assigned on an aircraft dispatch form. Agencies may have pre-assigned air-to-air frequencies for IA specific geographic areas. Specific radio frequencies should be assigned for extended attack incidents through the State Air Desk (Operations Officer – IMT) to prevent interference with other incidents.
- Each state uses different radio frequencies for communication between Leadplanes and Airtankers. In New South Wales, bombing frequencies are assigned based on the lead aircraft's home airbase.

Air-to-Air Continuity

The AAS and Air Attack pilot must monitor all assigned air-to-air frequencies and maintain communications with incident aircraft. Air resources under the direct supervision of the AAS or Leadplane must monitor their assigned air-to-air frequency.

---

[6] For operations outside of NSW, consult hosting dispatch center for local procedures.

[7] Western Australia uses a VHF-FM frequency known as LAT TAC in place of the VHF-AM LAT tactical frequency.

5.3.5 Conflicting Radio Frequencies

When multiple incidents in relative proximity are sharing the same tactical frequencies, interference can seriously impair operations. Aerial supervisors must recognise this and request different frequencies through the State Air Desk or Air Operations Officer during extended attack,

5.4 Incident Airspace; the Fire Traffic Area

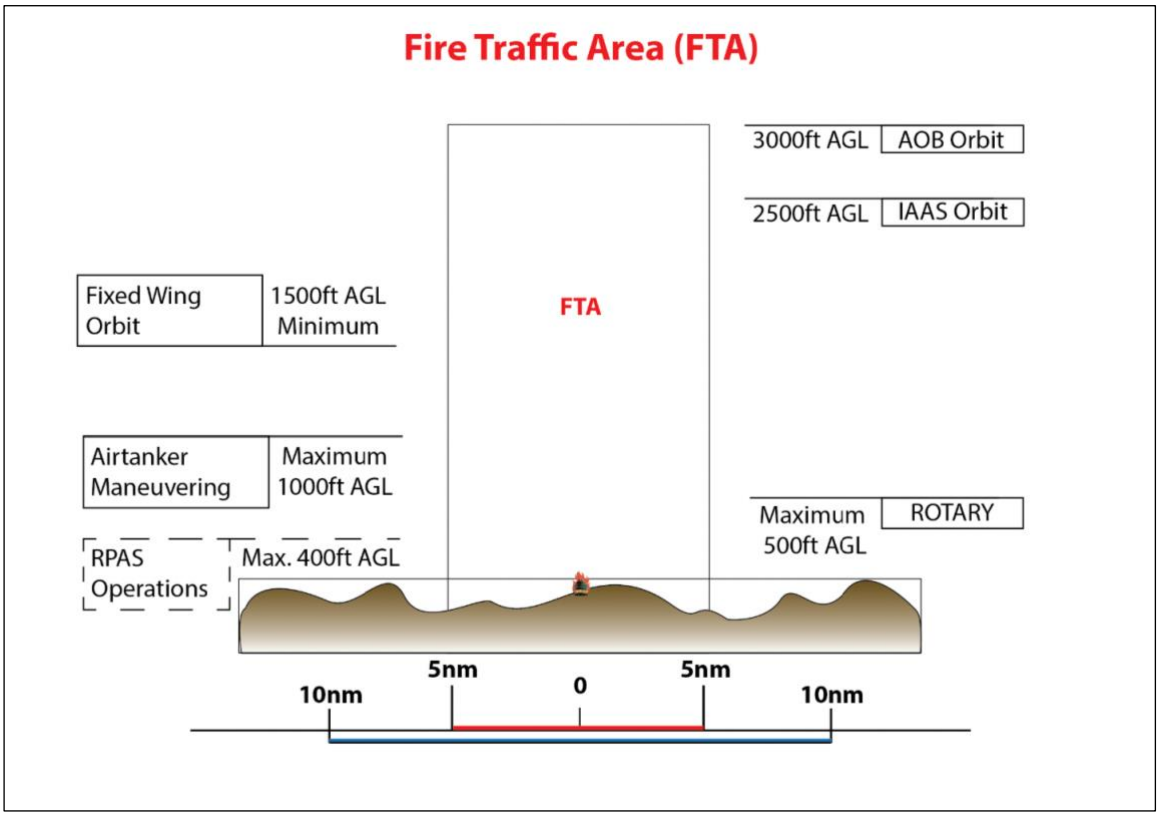
The airspace surrounding an incident is managed by the aerial supervisor who must implement FTA procedures. All wildland incidents, regardless of aircraft on scene, have an FTA. If aerial supervision is not on scene, the first aircraft on scene will establish the FTA protocol.

The FTA is a communication protocol for firefighting agencies. It does not pertain to non-participating aircraft.

5.4.1 FTA Description

The FTA is a section of airspace with a 5 Nautical Mile (NM) buffer from the incident and up to 3,000 feet Above Ground Level (AGL).

Figure 2. Fire Traffic Area



Key components and procedures of the FTA include:

- *Initial Communication Ring*—A ring 10 Nautical Miles (nm) from the centre point of the incident. At or prior to 10 nm, inbound aircraft contact the IAAS or appropriate aerial resource for permission to proceed to the incident. Briefing information is provided to the inbound aircraft by the aerial supervisor over the incident.

- *FTA Boundary*—5 nm from the centre point of the incident that must not be crossed by inbound aircraft without first receiving clearance from the appropriate on-scene incident aircraft.
- Three (3) Cs of initial contact – Communication requirements and related actions to be undertaken by the pilot of the inbound aircraft:
  - *Communication*—Establish communications with the controlling aerial supervisor or an on-scene aircraft if there is no aerial supervision.
  - *Clearance*—Receive clearance from aerial supervisor (or on-scene aircraft if there is no aerial supervision) to proceed to the incident past the FTA boundary. The inbound pilot will acknowledge receipt of clearance or (hold) outside the FTA until the clearance is received and understood.
  - *Comply*—Inbound aircraft will comply with clearance. If compliance cannot be accomplished, the inbound aircraft will remain outside the FTA until an amended clearance is received and understood.
- Departing Aircraft – Aircraft departing the FTA must follow assigned departure route and altitude. Aerial supervisors must deconflict routes for departing aircraft within the FTA.

6 Mission Procedures

6.1 Aircraft Preparation

The pilot is responsible for ensuring the aircraft is mission-ready each day.

Preflight Inspection:

- Conduct a thorough preflight inspection to verify the aircraft's airworthiness.
- Check critical systems, avionics, and emergency equipment.
- Ensure the aircraft is properly fuelled to meet mission requirements.
- Confirm that all maintenance discrepancies have been addressed and logged appropriately.

Mission Equipment:

- Verify the presence and functionality of all required mission-specific equipment, such as kneeboard and iPad.

6.2 Flight Risk Assessment Tool (FRAT)

See Section 3.2 for FRAT requirements and procedures.

6.3 Dispatch Procedures

6.3.1 Mission Tasking Requirements

The Airtanker Dispatch Form must be issued by the State Air Desk (SAD) and sent to the LAT Air Base Manager (ABM), LAT AAS, and the aircraft being dispatched.[8]

The form must contain all available incident information, including at a minimum:

- Latitude and Longitude – preferably in Degrees, Decimal Minutes
- Geographic Location – as identified on a map or chart.
- Incident AAS Contact Details.
- Communications Plan – including Firecom, Incident Control Centre (ICC) channel, FCTAF, and LAT tactical frequency as a minimum.
- Aircraft Assigned/Operating – details of aircraft currently allocated to, or operating on, the incident.
- Weather Conditions – actual and forecast.
- Fire Activity and Threats.
- Load Product – type of suppressant/retardant.
- Additional Safety Information – any other details that may enhance operational safety.

[8] Dispatch is not authorised until the crew has received the Airtanker Dispatch Form from the SAD and has completed a FRAT.

6.3.2 Mission Briefing

The Mission Briefing is conducted prior to each mission to ensure the pilot and AAS are fully prepared and aligned on mission-specific details. The briefing addresses the unique challenges of the upcoming operation and establishes a clear, shared plan.

- Mission Overview
  - Outline the mission objectives and expected outcomes.
  - Review fireground conditions, including terrain, fire behaviour, and traffic density, to establish situational awareness.
- Flight Risk Assessment Tool (FRAT) Review
  - Review and update the FRAT, confirming mission suitability.
  - If the FRAT indicates elevated risk, discuss mitigations or alternatives and ensure approvals are obtained as required.
- Roles and Responsibilities
  - Confirm the respective responsibilities of the Air Attack pilot and the AAS.
- Final Confirmation
  - Confirm mutual understanding of the mission plan.
  - Reaffirm that either the Air Attack pilot or AAS may call for a tactical pause or terminate the mission if safety concerns arise.

6.4 En Route Procedures

6.4.1 After Take Off

Air Attack Supervisor will establish flight following:

- Call sign
- Departure location
- Persons on board
- Fuel on board (hours)
- ETE
- Destination

Flight Following Call

Example	*"Richmond Base, FS-200 is off Richmond, 2 POB, 4 hours of fuel, 25 minutes to the Treeline Fire."*

The crew should monitor assigned fire frequencies while en route to the incident to build situational awareness, particularly the FCTAF. When feasible, monitoring the LAT tactical frequency during distant dispatches allows early coordination with the Airtanker.

6.4.2 Diversions

Higher priority incidents may require diversion of aircraft. A reassignment may be given through the SAD or ABM. Upon receiving a divert notice, the aerial supervisor must attain, at a minimum, the following information:

- Incident name
- Incident coordinates
- Air and ground contacts
- Assigned frequencies
- Assigned resources

6.5 Fire Traffic Area Entry Procedures

Aerial supervision personnel must comply with FTA entry procedures prior to reaching 10 NM from the incident. The AAS will make the required FTA entry and coordination calls on behalf of the aircraft.

Entry procedures vary depending on whether aerial supervision is already established, absent with other aircraft present, or if no aircraft are on scene. The Air Attack pilot should maintain awareness of these calls and be prepared to assist as required, ensuring safe and coordinated integration into the incident airspace.

6.5.1 Scenario 1: Aerial Supervision on Scene

- Change to incident frequencies.
- Make a 10-mile call to aerial supervision with location and altitude.
- Obtain clearance, including:
  - QNH setting
  - Assigned FTA entry altitude
  - Aerial Supervision altitude
  - Altitudes of other aircraft
  - Hazards
- Read back clearance and enter incident airspace as briefed.
- Visually acquire Aerial Supervision; report distance/clock reference when in sight.
- Receive tactical briefing and confirm handoff of any responsibilities.
- Coordinate with IC/ground personnel to verify incident objectives and priorities.

6.5.2 Scenario 2: Aerial Supervision is Not on Scene, but Other Aircraft are on Scene

- Change to incident frequencies.
- Make a 10-mile radio call with location and altitude.
- Obtain clearance, including:
  - QNH setting
  - FTA clearance altitude.
  - Altitudes and location of aircraft on scene
  - Hazards

- Read back the clearance and enter incident airspace as coordinated with on-scene aircraft.
- Identify all on-scene aircraft, establish radio contact, and deconflict traffic as required.
- Coordinate with IC/ground personnel to verify objectives and operational priorities.

6.5.3 Scenario 3: There Are No Aircraft on Scene

- Make 10-mile and 5-mile calls in the blind on the assigned FCTAF frequencies.
- Contact IC/ground personnel on the assigned air-to-ground frequency to confirm no aircraft are on scene.
- Proceed to the incident, maintaining at least 2,500 feet AGL and vigilant lookout for traffic.
- Coordinate with IC/ground personnel to verify objectives and operational priorities.

Blind Call

Example	*"Any aircraft over the Treeline Fire, FS-200 is 10 southwest, inbound at 6,500, any aircraft please advise on Fire CTAF 124.55."*

6.6 Entering Incident Airspace

Standard entry is 2,500 feet AGL in a right-hand orbit to allow the AAS to establish situational awareness, unless operational requirements dictate otherwise.

6.7 First Aerial Supervision on Scene – Immediate Responsibilities

- *Account for traffic*—Identify all on-scene aircraft, establish radio contact, and deconflict traffic as required.
- *Establish vertical structure*—Determine ground elevation and assign mission flight altitudes to set FTA altitudes for all incoming aircraft (helicopters, airtankers, relief aerial supervision, media, etc.) if not already established.
- *Assess and brief hazards*—Identify and communicate hazards such as powerlines, antennas, snags, terrain, thunderstorms, excessive winds, poor visibility, and potential airspace conflicts.
- *Confirm objectives*—Coordinate with IC/ground personnel to confirm incident objectives and operational priorities.
- *Define LAT working area*—Establish the LAT working area and specify routing and checkpoints for all Rotary-Wing aircraft operating within it.

6.8 Fixed-Wing Scripts

Standardised scripts are the foundation of communication between aerial supervision and pilots. Effective missions rely on pilots receiving clear, timely, and properly sequenced information. Consistent phrasing and cadence ensure clarity, reduces workload, and supports safe, effective, and efficient mission execution.

See Appendix A for additional information on Fixed-Wing scripts.

6.9 Standard Clearances and Commands

The following clearances and commands are used to ensure standardised communication, clarity, and shared understanding between the AAS and flight crews.

When an AAS is directly supervising the Airtanker, the following clearances are issued:

- **Cleared to Manoeuvre**
  - Authorises the Airtanker to leave orbit altitude and manoeuvre as required to position for the drop.
  - Issued by the AAS when directly supervising Airtanker.
- **Cleared Right**
  - Airtankers must first be cleared for right-hand traffic and descend to the designated manoeuvring altitude before entering the pattern.
- **Cleared to Drop**
  - Indicates the Airtanker is established on the run, the line is clear, and the drop may commence.
  - Issued by the AAS when directly supervising Airtanker; not required when working with a Leadplane.
- **Go Around**
  - A "Go Around" may be called by any participant—including aerial supervision, Airtanker, helicopter, or ground personnel—whenever safety or efficiency is compromised.
  - During a Go Around, the Airtanker will:
    - Re-establish on downwind, or
    - If the load has been released, exit the Fire Traffic Area in accordance with the exit brief.

6.10 Air Attack Flight Profiles

6.10.1 Orbit

The standard orbit is normally flown at approximately 2,500 feet AGL in a right-hand pattern. Airspeed should be relatively slow[9] and steady to improve endurance, reduce G-loading, and provide the AAS with a stable, unobstructed viewing platform. Avoid drift smoke whenever possible and remain clear of the smoke column at all times. Adjust orbit position as required to optimise visibility and maintain safe separation from other traffic.

6.10.2 Altitude and Separation

A minimum of 1,000 feet vertical must be maintained from all participating aircraft within the FTA. Operations below 1,500 feet AGL are prohibited. These altitude and spacing requirements ensure safe integration with other firefighting aircraft.

6.10.3 Minimum Airspeed

Airspeed during normal operations shall not be flown below minimum controllable airspeed one engine inoperative (V_{MCA}).

6.10.4 Manoeuvring

Bank angles of approximately 30° should be targeted and must not exceed 50°. Manoeuvres should be smooth, coordinated, and predictable to maintain comfort for the AAS and

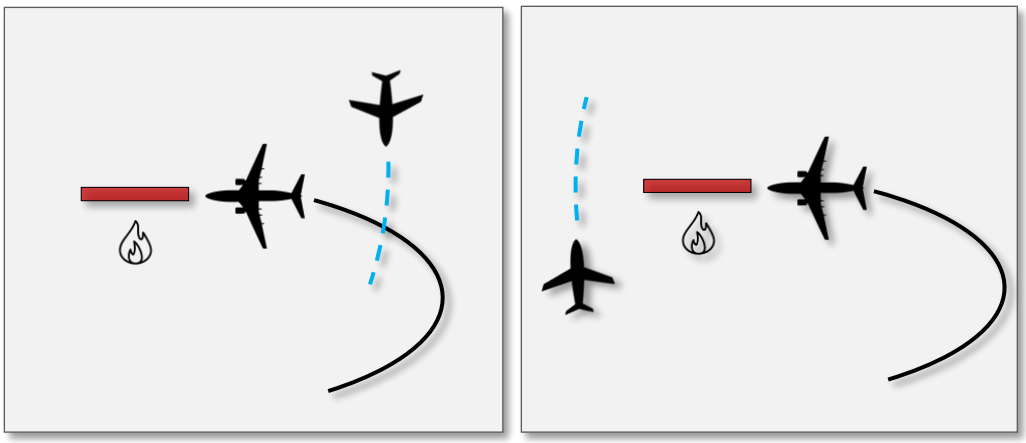
[9] Typically between 120 and 140 KIAS depending on aircraft type and conditions.

minimise pilot workload. Control inputs should be gradual to prevent abrupt attitude changes and preserve a stable observation environment throughout the orbit.

6.10.5 Tactical Positioning

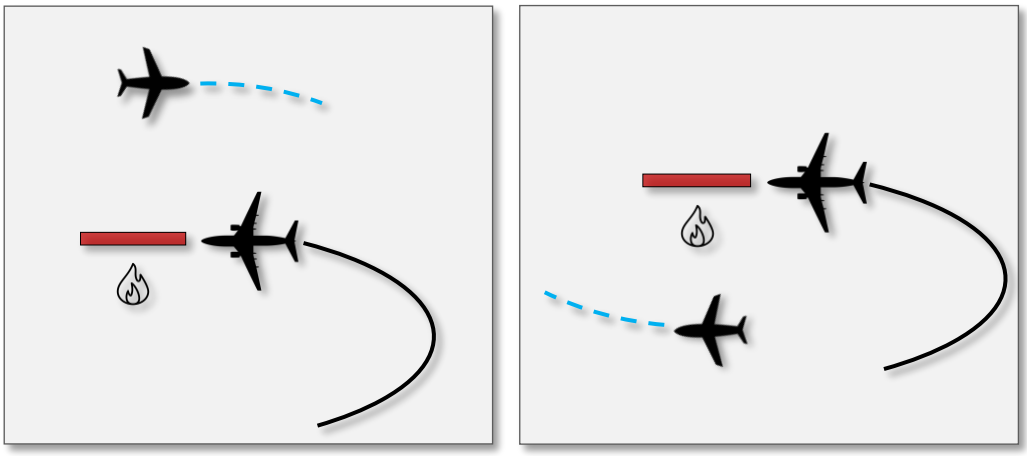
The Air Attack aircraft should be positioned to provide the AAS with an optimal vantage point for observing airtanker drops. The preferred position is directly behind the airtanker on the drop run, allowing full visibility of retardant fall and wind drift. An in-front position is an acceptable alternative when the preferred alignment cannot be maintained. Lateral spacing of approximately ¼ to ½ miles and a heading offset of roughly 90° from the airtanker's track provide an effective observation geometry while maintaining safe separation.

Diagram 2: Optimal Positioning



Positioning parallel to the airtanker's heading or abeam the drop path results in a reduced ability to assess retardant trajectory and wind drift. Such alignment limits the AAS's depth perception and situational awareness of the drop pattern and should be avoided whenever practicable. The aircraft should remain clear of the airtanker's ingress, egress, and drop path, and maintain altitude and separation parameters to preserve observation safety and effectiveness.

Diagram 3: Suboptimal Positioning



6.11 Maintaining Continuity of Supervision

Complex air operations or air operations involving a mix of air resources requires continuous supervision by an AAS or Leadplane. To maintain continuous supervision, the following procedures should be followed. Good planning will ensure continuity:

- Use Leadplane to fill gaps in IAAS coverage and manage air/ground operations in designated areas on complex incidents.
- Stagger aircraft refuelling so all aircraft are not down simultaneously.
- Monitor flight times. Anticipate the need for a relief AAS, Leadplane, or other air resources. Notify dispatcher or AOBD in a timely manner.
- Anticipate fuel needs.
- Recommend activation of portable reload bases to reduce turnaround time.
- Coordinate refuel and relief needs between aerial supervisors to ensure continuity of airspace management/supervision.

6.11.1 Relief Guidelines

Aerial supervision is mentally demanding. Long flight hours result in mental fatigue and reduced effectiveness. Consider the following staffing guidelines:

- If the aerial supervisor will fly more than 4 hours on any one flight, order a relief.
- On multi-day incidents, assign a second aerial supervisor and rotate.

6.12 Transition Briefings

The responsibility is on the current aerial supervisor to provide a quality transition briefing to the incoming aerial supervisor. Incoming aerial supervision should listen to assigned frequencies in route to gather situational awareness. During transition briefing, each pause should be acknowledged back with affirmation that the message is received.

6.12.1 Elements of a Transition Brief

- Frequencies – Confirm all assigned frequencies.
- Operational Objectives – Priorities (first, second, third…), chain of command.
- Fire Anatomy – Hazards, DIVCOM, dip sites, IP, checkpoints, routes, roads, helispots, retardant avoidance, etc.
- Resource – Aircraft, engines, crews, Airtanker bases, ground contacts, UAS, helibase, etc.
- Tasks – Point protection, aerial ignition, firing, direct/indirect, recons, repeater mission, etc.
- Questions – Open up for incoming AAS to ask questions.
- Aerial supervision relief times – Local time will be used, and time is estimated time of arrival (ETA) over the fire.
- Positive handoff – Incoming AAS assumes fire name air attack and takes all fire frequencies; outbound AAS goes back to tail number air attack and communicates transition complete and relief order to dispatch.

6.13 Non-Normal Situations

6.13.1 Airspace Conflicts and Intrusion into FTA

When incident airspace conflicts and intrusions occur, the aerial supervisor must:

- Immediately ensure the safety of incident aircraft.
- Notify incident aircraft in the immediate area of the position of the intruder.
- Attempt radio contact with intruder aircraft by use of VHF-AM (known Victor, local Unicom) frequencies.
- If radio contact can be established, request intruder departs FTA area (assign an altitude and heading if necessary). Request the intruder to stay in radio contact until clear of the FTA.
- If the aircraft is a legitimate "non-participating" aircraft and has the authority (law enforcement) to be within the area, communicate with the aircraft and advise incident aircraft of its presence. If possible, coordinate altitudes, and locations.
- If radio contact is not established do not attempt to drive, guide, or force the intruder from the area. The aerial supervisor must monitor intruder's position, altitude, and heading.

The aerial supervisor must ensure that incident aircraft are informed and kept clear of intruder. This may require removing incident aircraft and suspending operations for as long as intruder is considered a potential hazard.

6.13.2 Incident Within an Incident (IWI)

An Incident Within an Incident (IWI) is a secondary emergency occurring during wildfire operations, such as an injury, burn-over, vehicle accident, trapped equipment, or aircraft emergency, requiring immediate response alongside ongoing suppression activity.

When an In-Flight or Incident Within Incident (IWI) occurs, the Air Attack Supervisor (AAS) reports the event, communicates its location and nature, and notifies all participating aircraft. The AAS confirms each airtanker's remaining on-scene time and fuel state, designates a primary air-to-air coordinator, establishes a ground point of contact, and ensures all rotary-wing assets are informed and accounted for.

Operations are reassessed to determine whether suppression efforts should continue or be suspended.

- Non-essential aircraft may be held in orbit or released to base.
- Exclusion zones and revised routes are established to maintain safety and separation.
- If life or property is immediately threatened, available aircraft may be directed to the affected area until the threat is mitigated.
- Medical evacuation support should be coordinated promptly, with landing zones identified near the incident location.

Following the event, crews must document key details and actions, submit an SMS report, and complete all agency-required documentation to support post-incident review and continuous improvement.

6.13.3 Flight Emergencies for On-Scene Aircraft

When a flight emergency is declared, possibly as "Mayday, Mayday, Mayday" the aerial supervisor manages the emergency using appropriate procedures from the list below:

- Emergency is highest priority until aircraft lands safely.
- Determine pilot's intentions for managing situation.
- Clear the airspace for the pilot as needed.
- Dedicate and clear a frequency for the emergency.
- Jettison load if feasible.
- If problem persists, assist aircraft to return to base or alternate landing site.
- Alert incident medevac units.
- Prepare for suppression of a fire associated with an aircraft crash.
- Notify dispatch or airport tower for necessary crash/rescue protocol.

6.13.4 Incident Aircraft Mishap Considerations

When a mishap has occurred or an aircraft is missing, on-scene aerial supervision manages situation using appropriate procedures below:

- Consider ordering additional aerial supervision.
- Assign aircraft as needed to conduct search.
- Determine location. Check with dispatch for last known coordinates and direction of flight.
- Monitor emergency frequency (121.5) if crash site is not known or if the aircraft is missing and its status is unknown.
- Assign remaining aircraft to holding areas or return to base.
- Activate incident medevac plan through medical unit.
- Assign on-site aircraft and personnel to control aircraft fire and initiate life-saving measures if they can do so without jeopardising their safety.
- Advise IC/Operations.
- Consider suspending non-essential aircraft operations.
- Direct ground resources and air support operations to crash site.

6.13.5 Medevac of Incident Personnel

Consider the following as appropriate:

- Serve as a relay between accident site, helibase, and medical personnel.
- Determine accident site location – latitude and longitude.
- Obtain Medevac helicopter frequency – may be listed in Medevac Plan.[10]
- Assist rescue personnel with helispot location, etc.
- Provide helispot dust abatement with helicopter buckets as needed.
- Guide Medevac helicopter to accident site.

[10] IMTs and Incident Control Centres typically have established procedures for incidents within the incident.

7 Fire Traffic Area Management

7.1 Common Principles of Aircraft Separation

- Maintain standard VFR "see and avoid" principles
- Use the designated air-to-air frequency for position reporting
- Comply with established Fire Traffic Area (FTA) procedures

7.1.1 Aerial Supervisor's Role in Aircraft Separation

The Aerial Supervisor is responsible for:

- Structuring and briefing the incident airspace layout[11], ensuring all aircraft understand assigned altitudes, routes, and operating areas.
- Monitoring radio communications to maintain situational awareness, including:
  - Pilot-to-pilot position reports
  - Blind call position reports
  - Aircraft tracking and situational updates
  - Directional guidance or deconfliction instructions
  - Advisories on the position and heading of other aircraft

7.2 Vertical Separation

500 feet is the minimum vertical separation for missions in the same airspace. 1,000 feet is preferred and should be used whenever possible.

- Assign helicopters a hard ceiling (i.e., at or below 4,500 feet). Do not assign them 500 feet AGL or "low-level."
- Vertical stacking of Airtankers is discouraged. Utilise an orbit altitude racetrack pattern.
- It is common practice to put media helicopters above the IAAS to keep them away from firefighting aircraft.

Table 1– Standard Operational Altitudes and Patterns

Mission	Standard AGL (feet)	Standard Pattern
Media/Aerial Observer	3,500	Right or Left
AAS– Fixed-Wing	2,500	Right
AAS – Helicopter	500 to 2,500	Left or Right
Airtanker Orbit	1,500	Left
Airtanker Manoeuvring	150 to 1,000	Left
Leadplane	150 to 1,000	Left
Helicopters	0 to 500 (hard ceiling)	Left or Right

[11] The coordinates of the incident or Initial Point (IP) must be verified, updated, and communicated to dispatch to ensure inbound aircraft can determine appropriate contact or holding points if communications with controlling aircraft are not established.

7.3 Horizontal Separation

Aerial supervision must ensure there is adequate visibility to conduct operations safely regardless of the airspace classification.

Flight patterns must be adequate.

- Consult pilots before finalising patterns and routes.
- Advise pilots on location of other aircraft if visual contact has not been reported.
- FCTAF must be clear for pilots to give position reports.
- Geographic features, such as a ridge or a river, can be used as a virtual fence to separate aircraft provided aircraft maintain assigned flight patterns.
- No-fly zones must be established to ensure safe separation when simultaneous missions at the same elevation are within close proximity.
- *Below ridges*—For operations separated by a ridge, a "no-fly zone" 500 feet vertically below the ridge top can be established to ensure separation.
- *Near geographic dividing lines (virtual fence)*—If simultaneous operations near the dividing line conflict, a horizontal "no-fly zone" must be established, or missions must be sequenced to ensure adequate separation.

7.3.1 Virtual Fences

A virtual fence is a defined boundary used to delineate separate working areas and maintain safe separation between aircraft, including between helicopters or between helicopters and fixed-wing operations. It provides an effective means of managing airspace and reducing radio congestion on the air-to-air frequency.

Pilots may be required to report arrival at a virtual fence and wait for clearance from AAS before proceeding.

Geographic features that make effective checkpoints and virtual fences include:

- Roads
- Power lines
- Ridges
- Lakes

7.3.2 Helicopter Routes

Established point-to-point flightpaths for repetitive missions from helibase to helispots or sling sites, from dipsites to targets, etc. For safety, efficiency, and monitoring, the AAS, in consultation with the helibase manager and/or helicopter pilots, will ensure flight routes and communications procedures have been established and are known:

Helicopter Daisy Chains

Two or more helicopters can be assigned to the same targets and dipsites for repeated waterdrops. The AAS, in consultation with helicopter pilots, will establish a "daisy-chain" flight route for these operations ensuring helicopters maintain the same orbit direction and separation.

Helicopter Recon Flights

These flights can be difficult to monitor. Consider the following procedures to maintain safe separation of aircraft:

- Schedule recon flights during slow periods.
- Assign a specific route for the recon (clockwise, maintain assigned altitude).
- Establish Check Points and clearance protocol with recon aircraft.

7.3.3 IPs, Checkpoints, and Holding Areas

The aerial supervisor assigns incoming aircraft to non-conflicting airspaces, or holding areas, as needed. Coordinates or a geographic reference work best.

Initial Point (IP)

A fixed-wing reporting location clearly identified by the aerial supervisor. It may be a latitude/longitude or geographic point (landmark). IPs are used to route incoming aircraft to a known location before engaging in tactics.

Aircraft entering IPs will announce their direction of approach and intended destination via call in the blind script on the assigned primary air-to-air frequency.

Checkpoint

A helicopter reporting location clearly identified by the aerial supervisor. It may be a latitude/longitude or geographical point (landmark). Check points are used to route helicopter to and from assignments.

Helicopters using check points while transitioning an established route will announce their direction and intended destination via call in the blind script on the assigned air-to-air frequency. When sequencing aircraft helicopter check point should be set up close to the target area that is not in the fixed-wing final/departure flight path. Helicopters should call off each dip, drop.

Holding Areas

Any known location can be used by aerial supervisors to hold aircraft. There can be multiple areas on an incident being used at the same time for multiple aircraft at different locations.

- Pilots must be aware of other aircraft in their assigned holding area.
- Pilots must be able to communicate position reports to each other.
- Holding area must be clearly defined – by a geographic reference point or distance and direction relative to the incident aircraft will normally establish a “racetrack” pattern where they are flying at the same altitude and providing their own visual separation.
- Aircraft must receive clearance to depart the holding area once assigned.
- Helicopters can be held on the ground or in the air as needed to maintain adequate separation. Considerations include:
  - Pilots should be able to maintain forward flight rather than constant hover.
  - Long periods of holding helicopters should be done on the ground.

7.4 Working Area/Area of Operation (WA/AO)

WA/AOs are areas where an identified project or task is being accomplished i.e., (crew support, recon, logistics, retardant delivery, ground firing support, aerial ignition, troop shuttle, UAS, etc.) with one or more aircraft. Because incidents often have several active WA/AOs with different operating altitudes (based on terrain) aerial supervision resources assign routes, patterns, checkpoints, fences, initial points (IP), holding areas and altitudes to deconflict travel routes between WA/AOs and aircraft bases.

7.4.1 LAT Working Area

The LAT Working Area is a defined block of airspace established for the operation of Large Airtankers and supporting aircraft. It is managed by the LAT AAS in coordination with the Incident AAS.

The area generally extends from the surface to Orbit altitude and extends laterally 2–3 NM from the target. Boundaries and altitudes may be adjusted for terrain, visibility, or traffic.

Only aircraft coordinated by the Leadplane or LAT AAS may enter the LAT Working Area. Within it, the LAT AAS sequences aircraft, provides tactical guidance, and ensures safe separation from other aerial resources.

7.5 Active Sequencing

Sequencing is a technique used to deliver multiple aircraft to a shared target area. Sequencing can be done between fixed and helicopter aircrafts to the same target area but must be actively managed by the LAT AAS[12]. Aerial supervisors should establish an order and provide clearance for each aircraft to the target/drop area.

Checkpoints

Checkpoints are primarily used by helicopters and water scoopers and should be located near the target area. This allows aircraft to transit, deliver their load, and return to the dip site efficiently.

Checkpoints must never be located along the airtanker's base leg, final leg, or exit path.

Ideal helicopter checkpoint locations:

- Inside the airtanker drop pattern abeam the target area, allowing both aircraft to maintain visual contact throughout the manoeuvre.
- Abeam the target area on the opposite side of the airtanker pattern, providing mutual visibility without crossing flight paths.
- When terrain, smoke, or dip site locations prevent ideal placement, checkpoints may be positioned just outside the drop pattern, provided they are not in the airtanker's exit path. These positions increase transit time between the checkpoint and target area.

Routes and Circuits

Helicopters may transit freely between dip sites and checkpoints without aerial supervisor clearance. Therefore, routes between dip sites and checkpoints must not intersect the airtanker's flight path when the airtanker is below manoeuvring altitude or climbing away from the drop area.

Clearance to the Target Area

The order of clearance to the target depends on the airtanker's position in the drop pattern when the helicopter or scooper calls from the checkpoint:

- If the airtanker is descending from orbit or on early downwind, the helicopter is typically cleared first.
- If the airtanker is mid-downwind, the aerial supervisor determines sequence priority based on operational needs. Airtankers must remain flexible and prepared to extend or go around.

---

[12] Separation and wake turbulence minima vary between agencies and jurisdictions. Pilots should ensure compliance with local requirements and guidance before conducting sequencing operations.

- If the airtanker is late downwind, base, final, or exiting, the helicopter or scooper will normally be cleared as target number two.

This framework is adaptable to real-time conditions. At times, the airtanker may be asked to go around to accommodate a helicopter or scooper due to operational priorities; at others, the helicopter may hold to allow a retardant drop.

The overriding objective is to ensure the airtanker and helicopter are never at the same altitude and location simultaneously. Wake turbulence must also be considered when aircraft operate in close proximity.

Helicopter Clearances[13]

- *"Cleared to Target"*—Denotes a helicopter is cleared to a target/drop area.
- *"Cleared to Transition"*—Denotes a helicopter is cleared through the area (on way to helispot/slingspot, back to helibase, on a recon, etc.).
- *"Cleared Unrestricted"*—Denotes to a helicopter that the active sequencing has stopped and no longer a need to call for clearances at the designated checkpoint.

FCTAF Frequency Communications

LAT AAS	*"Helitak 206, call your dips, drops, and call for clearance at Rock Check"*
HLTK-206	*"Helitak 206 is off the dip"*
HLTK-206	*"Helitak 206 is at Rock Check"*
LAT AAS	*"Helitak 206, clear to target, number 2 behind B-210 on left base, caution wake turbulence"*
HLTK-206	*"Helitak 206 has visual on B-210, clear to target number 2."*

LAT Tactical Frequency Communications

AAS	*"B-210, cleared to drop, Helitak 206 is holding at your 2 o'clock."*

7.5.1 LAT AAS Responsibilities During Sequencing

The LAT AAS and Air Attack pilots use effective crew resource management to maintain clarity, safety, and operational tempo during aerial supervision and sequencing activities.

- Ensure critical transmissions are made on both the primary and secondary frequencies.
- Maintain full situational awareness of aircraft positions, fire behaviour, and airspace activity.
- Adjust operational tempo and tone as the situation evolves.

[13] Using this specific language enhances situational awareness for all aircraft operating within the FTA. Sequencing clearances should not be confused with a "cleared in" call, which authorises an aircraft to enter the FTA.

7.6 CTAF and LAT Tactical Frequency Management

The LAT AAS and Air Attack pilot must monitor both the FCTAF and Firebombing frequencies throughout the mission. The LAT tactical frequency is designated for fixed-wing operations.

7.7 Coordination Between Aerial Supervisors

Each incident is unique, and circumstances dictate that workload shifts between LAT AAS and IAAS as their responsibilities overlap in several areas. Operational continuity is achieved by briefing and positive hand off.

It is important that IAAS and LAT AAS work as a team and share workload commensurate with fire complexity, training, and position authority.

7.7.1 Positive Handoff of Aircraft

Anytime aircraft is handed off (whether requested or offered) to another aerial supervisor, both aerial supervisors first must agree to that handoff, and a positive hand off will occur. A positive three step process using call sign of those aircraft being handed off will be used.

Example

IAAS	*"FS-200, I'll hand off HLTK-206 and HLTK-207 for you to coordinate and sequence between Airtankers."*
LAT AAS	*"Roger. FS-200 has control of HLTK-206 and HLTK-207."*
IASS	*"Affirmative."*

8 Fire Behaviour and Fire Weather

An in-depth understanding of fire behaviour and fire weather is critical for Aerial firefighters, as these factors directly influence drop tactics, safety, and operational effectiveness. This section covers the primary elements affecting fire spread, including fuel characteristics, topography, and weather conditions, along with the diurnal cycle and other atmospheric influences.

8.1 Fundamentals of Fire Behaviour

Wildfire behaviour is shaped by the "fire triangle," which consists of three core components: fuel, weather, and topography. These elements interact dynamically, determining the intensity, spread, and unpredictability of wildfires.

8.1.1 Fuel

The type, quantity, and moisture content of fuel significantly impact fire behaviour.

- *Fuel Types*—The landscape contains a variety of fuel types that burn differently. Grasslands spread fires rapidly but with lower intensity, while dense eucalyptus forests, with their oil-rich leaves, can produce intense, fast-moving flames. Shrublands and forests with layered canopies often yield persistent, high-energy fires.
- *Fuel Moisture*—Fuel moisture varies with humidity, precipitation, and seasonal drying. Drier fuels ignite more readily and spread faster, while wetter fuels require more energy to combust. Fuel moisture is crucial in determining a fire's intensity and speed.

8.1.2 Weather

Weather is one of the most influential factors affecting fire behaviour, impacting both the likelihood of ignition and the spread rate.

- *Wind*—Wind is often the primary driver of fire spread, pushing flames toward unburned fuel, increasing oxygen supply, and even carrying embers to ignite new fires. Wind also increases fire intensity, creating erratic and challenging conditions for firefighting efforts.
- *Temperature and Humidity*—High temperatures and low humidity dry out fuels, making them more susceptible to ignition. Relative humidity has a direct effect on fuel moisture; as humidity decreases, fuels become drier and more combustible.
- *Precipitation*—Rain can reduce fire intensity by dampening fuels, though heavy or sheltered fuels may retain enough heat to reignite after light rain. Prolonged precipitation can diminish a fire's activity, but only for as long as fuels remain sufficiently wet.

8.1.3 Topography

Terrain plays a significant role in fire spread, particularly in areas with steep slopes or confined valleys.

- *Slope*—Fires spread faster uphill, as flames preheat the fuel above them. The steeper the slope, the quicker the fire ascends, complicating containment efforts in mountainous regions.
- *Aspect*—In the Southern Hemisphere, north-facing slopes receive more sunlight, drying out fuels and increasing ignition potential, especially in the afternoon when temperatures peak.
- *Natural Barriers*—Rivers, roads, and rocky terrain can serve as natural fire breaks. However, certain terrain features, such as ridges and valleys, may channel wind and accelerate fire spread, especially during strong wind events.

8.2 The Influence of Fire Weather

Fire weather refers to atmospheric conditions conducive to fire ignition and spread. This includes not only current weather but also patterns and cycles that shape fire behaviour over time.

Fire Danger Ratings (FDR)

In Australia, the Fire Danger Rating (FDR) system assesses potential fire severity based on weather, fuel conditions, and historical fire data. The FDR scale indicates risk levels from "Low-Moderate" to "Catastrophic," helping agencies determine readiness levels and deploy resources accordingly.

8.2.1 Temperature and Relative Humidity Patterns

- *Temperature*—Higher temperatures increase the likelihood of ignition and boost fire intensity by drying out fuels. Temperature often peaks in the afternoon, aligning with the most intense fire behaviour.
- *Relative Humidity*—Low relative humidity dries fuels quickly, making them more prone to ignition. As humidity decreases, fuels reach a critical moisture threshold, increasing fire activity. In arid conditions, fires may spread more aggressively, necessitating cautious aerial operations.

8.2.2 Wind Behaviour

Wind is a key factor in fire behaviour, influencing both direction and rate of spread.

- *Prevailing Winds*—Prevailing winds, such as coastal breezes, can exacerbate fire spread in certain areas and create patterns of spread that firefighters can predict and plan for.
- *Gusts and Erratic Winds*—Wind gusts can rapidly alter fire direction and intensity, challenging containment efforts. Erratic winds, often produced by terrain or localized weather, can create rapid shifts in fire behaviour, necessitating constant monitoring.
- *Foehn and Downslope Winds*—These dry, warm winds, such as Australia's "northwesterlies," reduce humidity, accelerate drying, and significantly increase fire danger, often leading to extreme fire behaviour.

8.2.3 Atmospheric Stability and the Diurnal Cycle

- *Stability and Instability*—Atmospheric stability affects fire behaviour. Unstable conditions, characterized by rising warm air, can enhance fire intensity and support vertical fire growth. Conversely, stable conditions inhibit vertical development and may slow fire spread.
- *Diurnal Cycle*—Fire behaviour changes over a 24-hour period. Typically, fire activity subsides at night and early morning when temperatures cool and humidity rises. Fire intensity tends to peak during the afternoon when temperatures are highest, and relative humidity is lowest.

8.2.4 Smoke Plumes and Convection

- *Convection Columns*—Large fires can produce convection columns, or "pyrocumulus" clouds, which lift smoke and embers high into the atmosphere. This can lead to spotting, where embers ignite new fires downwind from the main blaze.
- *Smoke Movement*—Wind direction and topography affect smoke dispersion. Smoke movement impacts visibility for both air and ground crews, sometimes limiting operational options or requiring tactical adjustments.

8.2.5 Spotting Potential and Ember Cast

Spotting occurs when embers are lofted ahead of the main fire by convection or wind, igniting new fires. Spotting potential is influenced by fuel type, fire intensity, and wind speed, making it a significant consideration in aerial operations. Fires in volatile fuels, such as eucalyptus forests, have high spotting potential, especially in windy conditions, and can rapidly increase the fire perimeter.

8.3 Types of Fire Activity

Different types of fire activity indicate the fire's intensity, spread pattern, and overall behaviour. Recognising these fire types is essential for assessing threats and determining appropriate tactical responses.

- *Running Fire*—A running fire moves rapidly across the landscape, driven by wind or slope. This type of fire spreads intensely and often produces long flame lengths, requiring aggressive containment strategies.
- *Crowning Fire*—In a crowning fire, flames advance through the canopy layer of trees. This type of fire behaviour is common in eucalyptus forests and can lead to intense, fast-moving fires. Crowning fires are difficult to contain due to their height and speed.
- *Spotting*—Spotting occurs when embers are carried by wind or convection currents, igniting new fires beyond the primary fire perimeter. Spotting complicates containment and can rapidly expand the fire's reach.
- *Backing Fire*—A backing fire spreads slowly downhill or against the wind. This type of fire is typically less intense but can still produce significant heat. Backing fires may allow for more controlled containment.
- *Flanking Fire*—Flanking fires spread along the sides of the fire, moving perpendicular to the wind or slope. These fires can grow the perimeter, posing containment challenges, particularly in high-wind conditions.
- *Torching*—Torching is the isolated burning of individual trees or shrubs, where flames travel from the ground up into the canopy. Torching can serve as an indicator of fire intensity and potential crowning behaviour if more widespread.

Fire behaviour and fire weather are dynamic, interconnected factors that influence every aspect of wildfire operations. By understanding how fuel, topography, and weather interact to shape fire spread—and by recognising various types of fire activity—Crewss can make informed decisions on drop placement, timing, and coordination with ground crews. Mastery of fire behaviour concepts enables Crewss to adapt to changing conditions, prioritise safety, and execute effective aerial tactics.

9 Aerial Fire Suppression Strategies

Principles that apply to ground operations also apply to air operations (anchor, flank, and pinch, etc.). Strategies are based on values at risk and resource management objectives, while tactics are based on fuel type, fire intensity, rate of spread, resource availability, and estimated line production rate.

Aerial supervisors make decisions based on objectives developed by incident command personnel. Aerial supervisors are obligated to assist the IC and Operations personnel with strategic advice during multiple ignition events and extended attack incidents relating to aviation resource capabilities and needs.

Note: Aerial application of suppressants and retardants should be used in support of ground resources support and be anchored.

9.1 Aerial Fire Suppression Strategies

Direct Attack

Drops directly adjacent to fire edge for retardant and directly on the fire edge for retardant, water, and suppressants, in support of ground forces. If you want retardant to land in or partially on the fire edge, it needs to be specified (half in/half out, etc.); conversely if you want water/suppressants to land directly adjacent to the fire edge, it needs to be specified (pretreat the green).

Parallel Attack

Parallel to and within a hundred feet of perimeter. Anticipate lateral fire spread, safety, and line construction rates of resources assigned. This is a common practice for retardant use when ground fuels are carrying the fire as it allows time to tag on and extend prior to individual drops being hooked around by the fire.

Indirect Attack

Used to enhance control lines established by ground forces in advance of the fire. Also used for structure defence and safety zones.

9.2 Aerial Fire Suppression Tactics

A single Airtanker frequently performs multiple drops to build a retardant line, form a “V” off the head or heel, or shape the fire’s behaviour. Choice of tactic depends on fire size, slope, wind, available aircraft, and the objective (containment, check head, protect asset). The Leadplane and LAT AAS must coordinate pattern, spacing, and coverage level for each tactic.

Anchor-and-Flank (Primary Line Construction)

Drop a line on an established anchor (natural or constructed) and continue driving the line away from the anchor to contain a flank or check progression. Best where there is a defensible anchor (road, creek, burnt ground) and sufficient airtanker load to build an effective line.

Boxing / Perimeter (Small Fires)

Multiple consecutive drops form a boxed perimeter around a small fire. Useful for initial attack when crews will mop up inside the boxed area. Requires rapid, repeated passes and good crew/ground coordination to ensure the safety of firefighters who are on the line.

V (Head or Heel Control)

Form a V that either pinches the head (V off the head) to redirect or slow the head, or a reverse-V at the heel. This is effective for containing a slop-over or creating an anchor at the heel.

Parallel / Stacking Pattern (Steep or Broken Ground)

When slopes prevent a boxed approach, contour drops along the slope in parallel lines or stacked passes from top to bottom. Start at the top and progress downhill to limit spot runs.

Full Coverage / Blanket Drop (Delayed Attack & Spot Control)

Blanket the entire fire area with multiple passes to reduce intensity rapidly. For small spot fires or delayed attack scenarios, use a heavy coverage. Partial loads can reduce risk of a miss and refine aim based on feedback.

Point Protection / Structure Defence

Place a concentrated strip or pad between the fire and critical asset to reduce flame contact and radiant heat. Work in coordination to support ground crews and do not assume retardant alone is sufficient.

Reinforcement (Support for Indirect Lines)

Apply retardant along the non-fire side of a pre-identified indirect control line (ridge or road) to reinforce a planned handline or containment anchor before ground attack reaches it.

9.3 IA and Multiple Fire Operations

9.3.1 IA Mission Priorities

During initial attack operations, aviation resources must operate in accordance with established FTA protocols. Aerial Supervision resources should consider the following responsibilities when an Incident Controller has not yet been established.

The Air Attack Supervisor, in consultation with dispatch, should:

- Conduct an initial fire size-up.
- Recommend resources based on observed fire behaviour, access, response time, resource availability, and capability.
- Develop a tactical plan to guide responding aircraft and ground resources.
- Provide periodic status updates to dispatch and responding resources.
- Assist inbound resources in locating the incident.
- Brief ground resources on potential safety concerns and expected fire behaviour.
- Assign arriving resources in accordance with the tactical plan until a qualified IC assumes command.

9.3.2 Multiple Fire Situations

An AAS may be activated during periods of multiple fire starts and will typically assist with the following:

Fire Detection

Identify new ignitions and report latitude/longitude coordinates, legal land descriptions, and other location details.

Establishing Incident Priorities

- Incident priorities should be based on:
- Threat to life and property
- Ownership and jurisdiction
- Current and expected fire behaviour
- Environmental sensitivity
- Political considerations
- Potential resource or asset loss

Determining Access

Assess access options including aerial insertion of personnel, road or trail access, and associated distance or time requirements.

Recommending Initial Attack Resources

Recommend resources based on capability, mode of access, probable availability, and response time.

Developing IA Strategy and Tactics

- Develop strategy and tactics aligned with resource objectives, expected fire behaviour, and the type and number of resources responding within specific time frames.
- Direct responding resources in accordance with the strategic and tactical plan until a qualified Incident Controller arrives.
- Provide regular intelligence updates to dispatch and the IC.
- Reassign resources to higher-priority incidents as required.

9.4 Delayed Attack Fires

When many small fires have started in a widespread area, resources are usually in short supply. An AAS may be assigned to assess and prioritise fires. Delayed attack fires, or fires that cannot be staffed within a few hours, may require a holding action until ground resources are available. Timely drops while the fire is small can be effective in holding or containing a fire temporarily. Retardant is much more effective than water.

During these situations the AAS will:

- Determine delayed attack fires requiring retardant. Request resources as needed.
- Set priorities. Consider flight time between fires. If priorities are equal, consider dropping on fires that are near each other before moving to fires some distance away.
- Direct retardant drops. Blanket covering of the entire fire is recommended when controlling both fire spread and fire intensity on small fires. While drops covering the fire reduce fire intensity, they also make burnout operations difficult if not impossible.
- Monitor status of fires and change priorities as necessary.

9.5 Wildland Urban Interface

9.5.1 Urban Interface Hazards

The following hazards to aircraft are often associated with urban interface incidents:

- Dense smoke and poor visibility.

- Power lines (may have to be de-energised).
- Antennas.
- Tall buildings.
- Media aircraft.
- Propane tanks.

9.5.2 Ground Safety

Urban interface incidents often have many citizens and homeowners scattered through the operations area. This can seriously impair tactical air operations and expose ground personnel to additional risk.

9.5.3 Effectiveness of Resources

It is critical that Airtanker and helicopter drops must be closely supervised to prevent inadvertent drops on non-incident persons and unnecessary damage to improvements. The aerial supervisor is responsible for providing the best available resources that can:

- Minimise risk to people and improvements.
- Provide assignments to aircraft which have increased manoeuvrability, drop accuracy, and quick turnaround times to targets.
- Drops are generally not effective on structures that are burning beyond the initial start phase.

9.5.4 Urban Interface Tactical Planning Principles

Apply the following principles in developing the tactical plan for air resources:

- Assess the situation and identify the following:
  - Identify air operational hazards.
  - Locate non-incident people in operations area.
  - Protection of evacuation routes.
  - Triage structures.
  - Identify possible dipsites and portable retardant plant sites.
  - Determine how air resources can best support suppression objectives.
- Request electrical transmission lines are de-energised. Don’t assume that they will be. Warn ground personnel not to be under or near power lines during drops.
- Determine where Airtankers or helicopters can be most effective.
- Use Airtankers in areas where visibility, hazards, flight routes, and target selection ensure reasonable effectiveness and acceptable risk.
- Use helicopters on targets requiring more manoeuvrability and accuracy.
- When possible, avoid holding patterns with Airtankers overpopulated areas.

9.6 Fire Chemicals

9.6.1 Suppressants (Direct Attack)

A fire suppression chemical applied directly to the flame base to extinguish the flame (water, foam, gel/water enhancer).

Foam Fire Suppressants

Foam fire suppressants contain foaming and wetting agents. The foaming agents and percentage of concentrate added to water affect the accuracy of an aerial drop, how fast the water drains from the foam, and how well the product clings to the fuel surfaces. The wetting agents increase the ability of the drained water to penetrate fuels. These products are dependent on the water they contain to suppress the fire. Once the water they contain has evaporated, they are no longer effective. Engines, portable pumps, helicopters, and SEATs may apply foam. Some agencies also allow the application of foam from fixed- wing water scoopers.

Water Enhancers

Water enhancers contain ingredients designed to alter the physical characteristics of water to increase viscosity, accuracy of the drop, or adhesion to fuels. They improve the ability of water to cling to vertical and smooth surfaces. The consistency of these products can change depending on the quality of the water

9.6.2 Long-Term Retardant (Indirect and Direct Attack)

Retardants contain fertiliser salts that change the way fuels burn. They are effective even after the water has evaporated. Large Airtankers, SEATs, helicopter buckets, and ground engines may apply retardant. Some retardant products are approved for fixed-tank helicopters. See the USFS Quality Products List for specific uses for each product.

- Recommended coverage levels and guidelines for use can be found in the *IRPG*, under Principles for Airtanker and Water Scooper Use.
- Retardant mixing, blending, testing, and sampling requirements can be found at the Forest Service (FS) WFSC website.[14]

9.6.3 Duration

In general, one can expect chemicals to remain effective for the following amounts of time[15]:

- *Long-Term Retardants*—Days to weeks (or until removed by environmental elements such as rain or wind).
- *Foams*—Minutes.
- *Water Enhancers/Gels*—Minutes up to possibly an hour or more (direct sunlight breaks down gels faster).

Note: "Refreshing" dried water enhancers with waterdrops do not provide any additional effectiveness than the water being dropped.

9.7 Coverage Level Section

Coverage level selection must also account for aircraft performance and environmental factors. Drop height and groundspeed directly influence the effective coverage on the target. Drops conducted with a strong tailwind component, or at high density altitude, result in increased groundspeed and a wider, lighter retardant pattern. In these conditions, higher coverage levels may be warranted to achieve the desired effect on the ground.

Time of day and relative humidity also influence coverage effectiveness. Toward evening, as solar heating decreases and relative humidity rises, fire behaviour typically moderates. Under

[14] Forest Service (FS) WFSC
[15] Time will vary according to weather conditions (heat, humidity, wind, etc.).

these conditions, a lighter coverage level may be sufficient to hold, whereas the same coverage may be ineffective during the peak burning period of the day.

Table 3. Recommended Coverage Levels

Coverage Level	Fuel Model Description	Typical Fire Behaviour
CL-1	Short/medium grass, spinifex	Creeping or smouldering; minimal flame front
CL-2	Grass with scattered trees	Slow-moving surface fire
CL-3	Mallee or low heath	Backing or flanking fire; steady, low flame front
CL-4	Open dry eucalypt forest	Running surface fire; moderate flame lengths
CL-6	Mixed forest with dense mid-storey	Well-developed running fire; intermittent torching
CL-8	Dense scrub, Mountain/alpine ash forest	Vigorous running fire: frequent torching, short crown runs

9.8 Ten Principles of Retardant Application

1. Determine the strategy; direct, parallel, or indirect, based on fire size up and resources available.
2. Establish an anchor point and work from it.
3. Use the proper drop height.
4. Apply proper coverage levels.
5. Drop downhill, with the sun behind the aircraft, when feasible.
6. Drop into the wind for best accuracy.
7. Maintain honest evaluation and effective communication between the ground and air.
8. Use direct attack when ground support is available, or it is feasible to extinguish the fire.
9. Plan drops so that they can be extended or intersected effectively.
10. Monitor effectiveness of retardant and adjust use accordingly.

Specifics on these principles are explained below.

- Airtanker pilots are responsible for the placement of their retardant/water load. Dropping on people must be avoided.
- Fly the appropriate drop height over the target. Minimum drop height is delineated in feet above the top of the vegetation. Minimum drop height is predicated on dropping at correct drop speeds.
- When in the Lead profile, the Leadplane should fly an offset line to compensate for crosswind drift, improving retardant placement accuracy for the Airtanker.
- Retardant should lose forward momentum and rain straight down for best coverage. For optimum retardant effectiveness, the forward motion must be completely stopped prior to the top of vegetation. This prevents shadowing where retardant with forward motion coats mainly the front of vegetation or other object creating an area of low or no coverage behind the object.
- Drop at appropriate coverage level. Airtanker pilots should drop at the coverage level requested by aerial supervision. Time permitting, a pilot that believes the coverage level should be adjusted may query the aerial supervisor or make a recommendation. This is especially appropriate if the drop will need to be executed at a higher-than-normal altitude or airspeed. The coverage level should be based on vegetation/fuel model. Observed fire

behaviour, terrain, or the effectiveness of the previous retardant drops may dictate changes from the standard coverage levels for given fuel model. In any case, if the airtanker drops at a different level than requested, that needs to be communicated to the supervisor.

- Strong Tie In: Pilots must be familiar with the drop patterns, limitations, and volume of the tank system in the Airtanker. When an Airtanker is directed to "tie in to" or "tag and extend" existing retardant with a specific coverage level, the airtanker pilot is expected to start the drop to provide the proper overlap at the directed start point. For a tag and extend of the existing retardant, the pilot should start their drop to ensure that the retardant line is a continuous line at the requested coverage level. Aerial Supervisor will normally communicate the start point as the end of the existing retardant with the understanding that airtanker pilots will push the release button at the correct time to provide proper overlap. A weak tie may be worse than a clear gap as it can be extremely difficult to observe from the air and may not be fixed, resulting in a failure of the retardant line.
- Plan drops so they may be extended or intersected effectively. Pilots should plan drops so they may be effectively extended/tied into or intersected. The "tail" length and location should be based on the estimated time until the next airtanker drop. For example, if extending the line along the flank of a fire and the next airtanker to drop is already in orbit, the end of the first drop may be very direct as the next drop will occur in a few minutes and before the fire has a chance to significantly grow laterally. If the next airtanker will not be arriving for 20 minutes, the tail must be more indirect so that the fire does not "hook" around the end of the first drop prior to the next airtanker arriving. Similar concepts apply when putting a "vee" across the head of fast-moving fire or when the retardant line will change direction.

10 Situational Awareness and Decision-Making

10.1 Maintaining Situational Awareness

Situational awareness involves developing a comprehensive understanding of the operational area, including the Fire Traffic Area (FTA), aircraft movements, and environmental factors. This broad awareness enables aerial supervisors to anticipate challenges and adjust plans quickly in response to new information. As the first line of defence for Airtanker safety, the Leadplane serves as a safety "relief valve," tasked with identifying and communicating any risks or deteriorating conditions in a timely manner.

10.1.1 Tracking Aircraft Positioning

Maintaining awareness of all aircraft positions within the FTA is essential for avoiding conflicts and coordinating operations effectively.

- *Visual and Radio Updates*—Regular visual and radio updates with other aircraft, especially the Airtanker and helicopters, ensure alignment with the tactical plan and prevent conflicts. Pilots exchange information to confirm relative positions and intended manoeuvres.
- *Airspace Coordination*—Understanding the layout of the FTA, including designated ingress and egress points and altitude layers, helps the Leadplane manage airspace effectively. This organisation reduces the risk of airspace conflicts and enables efficient sequencing.

10.1.2 Monitoring Environmental Changes

Conditions such as fire behaviour, wind shifts, visibility, and temperature can change rapidly, impacting flight safety and mission success.

- *Environmental Indicators*—Observing smoke patterns, atmospheric stability, and wind shifts provides the aerial supervisor with cues to anticipate changes in fire behaviour, assessing the suitability of drop paths and flight manoeuvres.
- *Hazard Awareness*—Maintaining an updated mental catalogue of known hazards—such as powerlines, ridgelines, vegetation, and other obstacles—ensures the aircraft operating in the low level environment can adjust flight paths as necessary. This proactive hazard awareness supports safe operations and minimises unexpected risks.

10.1.3 Assessing Fire Behaviour

Situational awareness includes closely observing fire behaviour, as it can influence tactical decisions and resource allocation.

- *Evaluating Spread and Intensity*—Regular assessments of fire behaviour, such as spotting, crowning, and rate of spread, allow the aerial supervisor to adjust tactics in alignment with the Incident Controller's (IC) objectives.
- *Proactive Risk Communication*—The aerial supervisor is responsible for identifying any deteriorating situations or emerging risks, promptly communicating these to ground and air resources. This timely communication serves as an early warning system to ensure adjustments can be made to maintain the safety and effectiveness of the mission.

10.2 Watch Out Situations

10.2.1 Low Light / Low Sun Angle

Reduced contrast, glare, and shadowing during low light or low sun angles can obscure terrain, smoke layers, or other aircraft. Pilots must increase visual scan and avoid operations in conditions where terrain definition or visibility of the drop zone becomes marginal.

10.2.2 Wake Turbulence

Wake turbulence from Large Airtankers and other aircraft can be significant and unpredictable, particularly in light wind or stable air. The LAT Air Attack aircraft operates above the Airtanker in the Fire Traffic Area (FTA) but must remain aware of airtanker entry and exit routes. Avoid crossing directly behind or below airtankers during these phases and maintain sufficient lateral or vertical spacing to prevent wake encounters.

10.2.3 Target Fixation

Focusing solely on the target can lead to loss of situational awareness or inadequate terrain clearance. Pilots must maintain disciplined scan patterns and cross-check altitude, attitude, and escape routes throughout the drop sequence. Tactical effectiveness must never override safety-of-flight.

10.2.4 Span of Control Exceeded

When the number of aircraft or mission complexity exceeds manageable limits, communication, separation, and safety margins can degrade rapidly. The Air Attack pilot should remain alert for signs of AAS workload saturation and, when necessary, recommend reducing the operational tempo or requesting additional aerial supervision resources before workload or safety limits are reached.

10.2.5 Mission Creep

Extending mission objectives beyond the original scope or pushing limits late in the operational period increases risk. Crews must recognise when fatigue, reduced visibility, or environmental changes are eroding margins, and should recommend termination or reassessment when conditions no longer support safe operations.

10.3 Judgment in High-Risk Environments

Sound judgment is essential in the unpredictable conditions of aerial firefighting. Pilots must evaluate risks, prioritise safety, and adapt to changing scenarios while making decisions that support mission objectives.

10.3.1 Prioritising Safety

Safety is the primary concern in high-risk firefighting environments, where changing conditions can pose immediate threats.

- *Risk Assessment*—The aerial supervisors continually evaluates risks related to fire intensity, environmental conditions, and proximity to obstacles. This thorough risk assessment informs each aspect of mission planning, from drop paths to entry and exit strategies.
- *Go/No-Go Decision-Making*—When conditions become unsafe, the pilot has the authority to call off the mission or exit the area. This judgment prioritises crew safety, ensuring that the mission proceeds only under safe conditions

10.3.2 Adaptability in Real-Time

Situational judgment requires flexibility, as the aerial supervisor may need to adjust plans quickly in response to changing conditions.

- *Dynamic Decision-Making*—Crews must assess and respond to real-time factors, such as sudden weather changes, evolving fire patterns, and unforeseen hazards. This may involve altering drop angles, adjusting flight paths, or coordinating with additional resources to maintain safe and effective operations.
- *Clear Communication*—When a decision impacts other resources, the pilot promptly communicates with the AAS, IC, and other involved parties. Effective communication promotes coordinated responses, reducing the risk of miscommunication in high-pressure situations.

10.4 Dynamic Risk Assessment During Operations

Dynamic risk assessment frames all operational decisions in the working area. Conditions can change within seconds, so decisions must adapt in real time. The Leadplane continuously evaluates terrain, weather, visibility, fire behaviour, traffic density, and aircraft performance to balance mission objectives with safety.

Dynamic risk assessment provides two immediate controls when risk exceeds acceptable limits. The first is a tactical pause to restore situational awareness and confirm controls. The second is mission termination when risk cannot be reduced to an acceptable level.

10.4.1 Mission Termination

If risk exceeds acceptable limits or conditions become unsafe to continue, the mission must be terminated.

- Any crew member who identifies a significant safety concern has authority to terminate.
- After termination, file an SMS report promptly that documents the decision, the underlying hazards, and recommended mitigations for future operations.
- Coordinate an orderly exit from the Fire Traffic Area in accordance with the exit brief.

10.5 The Eighteen Watch Out Situations

The Eighteen Watch Out Situations, published by the U.S. Forest Service under the NWCG, identify conditions that increase risk for ground firefighters. For aerial supervisors and pilots, awareness of these situations helps recognise when crews may be approaching operational limits, enabling better anticipation, communication, and tactical alignment to maintain safety across the incident.

1. Fire not scouted and sized up.
2. In country not seen in daylight.
3. Safety zones and escape routes not identified.
4. Unfamiliar with weather and local factors influencing fire behaviour.
5. Uninformed on strategy, tactics, and hazards.
6. Instructions and assignments not clear.
7. No communication link with crew members or supervisor.
8. Constructing line without a safe anchor point.
9. Building fireline downhill with fire below.
10. Attempting a frontal assault on the fire.
11. Unburned fuel between you and the fire.
12. Cannot see main fire, not in contact with someone who can.
13. On a hillside where rolling material can ignite fuel below.
14. Weather becoming hotter and drier.
15. Wind increases and/or changes direction.
16. Getting frequent spot fires across the line.
17. Terrain and fuels make escape to safety zones difficult.
18. Taking a nap near the fire line.

11 Post-Mission Responsibilities

The completion of a mission marks the beginning of a critical phase: debriefing, administrative follow-up, and documentation. These tasks ensure that lessons are learned, performance is assessed, and operational readiness is maintained.

11.1 After-Action Review (AAR)

Air Attack pilots should participate in After-Action Reviews (AARs) when conducted following a mission. These reviews provide valuable insight into mission outcomes, communication effectiveness, and overall coordination, contributing to continuous improvement across the program.

When present, the Air Attack pilot's input supports operational understanding from a flight operations perspective and helps reinforce coordination between aircrew and agency personnel. Participation should be viewed as an opportunity to share observations, support learning, and strengthen the team's collective performance.

11.2 Administrative Duties

The pilot is responsible for completing administrative duties related to the aircraft and mission. These tasks include:

- *Aircraft Logbooks*—Accurately recording flight times, maintenance events, and any operational notes.
- *Maintenance Discrepancies*—Logging and communicating any discrepancies identified during the mission and ensuring timely coordination with maintenance personnel.
- *Refuelling and Fuel Receipts*—Overseeing aircraft refuelling and ensuring fuel receipts are recorded and submitted as required.
- *Daily Reporting*—Completing and submitting daily operational reports as required by company policy.

11.3 Safety Reporting

The pilot is responsible for reporting any safety-related occurrences that may have occurred during the mission, including incidents, hazards, or near-misses. Reports should be thorough, accurate, and submitted in a timely manner in accordance with company policy.

Safety reporting reinforces a culture of accountability and continuous improvement in aerial firefighting operations.

Table of Contents

1 APPENDIX A - FIXED-WING SCRIPTS 1

1.1 FLIGHT FOLLOWING 1
1.2 10 NM INITIAL CALL 1
1.3 BLIND CALL 2
1.4 INITIAL CLEARANCE 2
1.5 TACTICAL BRIEF 2
1.6 CLEARANCE TO MANOEUVRE 3
1.7 DEPARTURE BRIEF 3
1.8 AIRTANKER BASE 3

1 Appendix A – Fixed-Wing Scripts

The following communication scripts are provided for reference only. In the Air Attack environment, all agency and tactical radio transmissions will be made by the Air Attack Supervisor (AAS). The purpose of this section is to familiarise Air Attack pilots with the structure, flow, and intent of standard aerial firefighting communications so they can maintain full situational awareness and understand the context of radio exchanges occurring during operations.

These scripts are based on the standardised, formatted communication methods used in aerial firefighting operations. The procedures have been developed and refined over many years to enhance clarity, predictability, and safety during high-workload environments.

It is recognised that terminology and communication styles differ between agencies across Australia, and not all personnel will use this format. These examples are not intended to replace the Air Attack Supervisor's communication methods or dictate how calls are made, but rather to provide structure and context for Coulson flight crews.

1.1 Flight Following

Initial Call – Required Items	LAT AAS
Dispatch Centre/Call sign	*"Richmond Base, FS-200* *off Richmond* *2 on board* *4 hours of fuel* *30-minute ETE* *to the Treeline Fire"*
Departure Location	
Number on board	
Fuel of board (hours)	
ETE	
Destination	

1.2 10 NM Initial Call

When responding to an incident with Aerial Supervision:

Required Items	LAT AAS
Aerial Supervisor call sign, your call sign	*"Treeline Air Attack, FS-200* *10 miles southeast* *inbound* *at 5000"*
Location	
Intent	
Altitude	

If clearance is given, continue at 170 kts or less to enter the Fire Traffic Area (FTA).

Comply with FTA entry instructions. If no contact:

- Hold outside 5 nm.
- Recheck assigned frequencies/volumes.
- Monitor air to ground for potential communication between participating aircraft and ground resources.
- Confirm with Dispatch Centre/Ground Resources if any other aircraft are on scene.
- If no other aircraft, enter the FTA as first on scene at the AAS Orbit altitude.

1.3 Blind Call

When responding to an incident without Aerial Supervision or status unknown:

Required Items	LAT AAS
Incident Name	*"Any aircraft over the Treeline Fire, FS-200 is 10 southwest, inbound at 6,500, any aircraft please advise on Fire CTAF 124.55."*
Airtanker Location	
Intent	
Altitude	
Frequency	

- Make 10-mile and 5-mile calls in the blind on the assigned FCTAF frequencies.
- Contact IC/ground personnel on the assigned air-to-ground frequency to confirm no aircraft are on scene.
- Proceed to the incident at Orbit altitude and maintain vigilant lookout for traffic.
- Coordinate with IC/ground personnel to verify objectives and operational priorities.

1.4 Initial Clearance

Required Items	LAT AAS	Airtanker
QHN setting	*"B-210, QNH 1014 Cleared in 3000 Air Attack is at 4000 helicopters at or below 2000 caution, towers and powerlines"*	*"B-210, 1014 cleared in 3000"*
Clearance Altitude		
AAS Altitude		
Other aircraft altitudes		
Hazards		

1.5 Tactical Brief

May be given in a variety of ways. Ensure briefing contains the following information:

Type	Details	LAT AAS
Orientation	Orientates pilot to specific areas of the fire. May be brief if returning to the same fire.	*"Your drop will be on the left flank of the fire, working the same area as before."*
Objectives	Tag and extend, direct, indirect, pretreat, point protection, etc. What's to be accomplished and why?	*"Objective is direct line, tag, and extend the line."*
Target Description	A standard target description includes the following: • Target Location • Drop objectives/Type of drop • Coverage level • Specific Hazards • Portion of load	*"Coverage level eight, full load, exit straight out, caution power lines along the road."*

1.6 Clearance to Manoeuvre

LAT AAS[16]	Airtanker Response Standard Pattern	Airtanker Response Right Pattern
"Clear To Manoeuvre" *or* *"Clear to Manoeuvre, Right Pattern"*	*"B-210, cleared to manoeuvre."*	*"B-210, cleared to manoeuvre right pattern."*
*	*"B-210 downwind"*	*"B-210 right downwind"*
*	*"B-210 base"*	*"B-210 right base"*
*	*"B-210 final"*	*"B-210 final"*
"I have your final, clear to drop."		

Go-Arounds/Split Loads:

A go-around may be communicated by anyone for safety/efficiency. During a go-around/split load the Airtanker should:

- Re-establish downwind at manoeuvring altitude, or:
- If load was released, exit the FTA at manoeuvring altitude or as briefed.
  - "B-210, off and empty" or "B-210, off the drop, half load remaining."

1.7 Departure Brief

Briefing Items	LAT AAS
Drop Evaluation – Start, Line, End	*"B-210, half load late, off one wingspan right load and return Richmond maintain 5000' exiting the FTA.'*
Load and Return/Hold/Release Location Special Instructions	

1.8 Airtanker Base

Approximately ten minutes from base, communicate the following:

Items	LAT AAS
Tanker Base/Call Sign	*"Richmond Base, FS-200* *10 minutes out* *Positive fuel"*
ETE	
Fuel Status	

[16] Clearance is required to maneuver and to drop. AAS is not required to acknowledge downwind and base legs.